

**CS-601 Update Book Important Mcq's**  
**For Final Term !!**  
**Solve By Vu-Topper RM!!**

وَتَعَزُّ مِنْ تَشَاءٍ وَتَذُلُّ مِنْ تَشَاءٍ



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 **0322-4021365**

**Question No:1**

**(Marks:1)**

**Vu-Topper RM**

Which one of the following is a Random-Access protocol?

TDMA

CDMA/CD

**ALOHA**

CDMA/CA

**Question No:2**

**(Marks:1)**

**Vu-Topper RM**

Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol.

**Transport**

Application

Session

**Question No:3**

**(Marks:1)**

**Vu-Topper RM**

In Block coding scheme, number of code words is always \_\_\_\_\_ data words.

Less than

Equal to

**Greater than**

Same as

**Question No:4**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_, each station sends a frame whenever it has a frame to send.

Slotted ALOHA

**Pure ALOHA**

None of the given

**Question No:5**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is/are un-reliable protocol(s).

IP and TCP

**IP and UDP**

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**Question No:6**

**(Marks:1)**

**Vu-Topper RM**

The extra bits added with the original data for error detection/correction are called \_\_\_\_\_.

Fault bits

Port numbers

**Redundant bits**

Process IDs

**Question No:7**

**(Marks:1)**

**Vu-Topper RM**

High-Level Data Link Control (HDLC) defines \_\_\_\_\_ types of frames to deal with the transfer modes.

One

**Two**

Three

Four

**Question No:8**

**(Marks:1)**

**Vu-Topper RM**

S-frames in High-Level Data Link Control (HDLC) are only used to transfer \_\_\_\_\_

User data

**Control**

Redundant

Original

**Question No:9**

**(Marks:1)**

**Vu-Topper RM**

The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

Duplicated

Blocked

Cleaned

**Escaped**

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**Question No:10**

**(Marks:1)**

**Vu-Topper RM**

Error correction is more \_\_\_\_\_ than the error detection.

Easy

Useless

Informal

**Difficult**

**Question No:11**

**(Marks:1)**

**Vu-Topper RM**

In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

**Polling**

Reservation

Token Passing

Port Forwarding

**Question No:12**

**(Marks:1)**

**Vu-Topper RM**

The type of ALOHA which improves the efficiency of Pure ALOHA is

**Slotted ALOHA**

Upper ALOHA

Integrated ALOHA

Lower ALOHA

**Question No:13**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is the process of converting digital data to digital signals.

CSMA/CD

CSMA/CA

**Line coding**

Segmentation

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**Question No:14**

**(Marks:1)**

**Vu-Topper RM**

Congestion Control is a feature of layer(s).

**Data Link**

Network

Transport

All mentioned

**Question No:15**

**(Marks:1)**

**Vu-Topper RM**

Collisions in CSMA/CA are avoided through \_\_\_\_\_.

The Interframe Space

The Contention Window

Acknowledgements

**All of the given**

**Question No:16**

**(Marks:1)**

**Vu-Topper RM**

Error detection and correction are the services provided by \_\_\_\_\_ layer.

**Data link layer**

Application layer

Physical layer

Session layer

**Question No:17**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is not a function of Data Link Control.

Framing

**Modulation**

Error Control

Flow Control

**Question No:18**

**(Marks:1)**

**Vu-Topper RM**

If the Hamming distance between sent and received code word is \_\_\_\_\_, then it shows that received data is corrupted.

**≠ 0**

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**Question No:19**

**(Marks:1)**

**Vu-Topper RM**

In even parity check code, the value of syndrome is \_\_\_\_\_ if the number of 1s is even.

1

2

**0**

4

**Question No:20**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_, each station is allocated a time slot during which it can send data.

CDMA

NDMA

**TDMA**

FDMA

**Question No:21**

**(Marks:1)**

**Vu-Topper RM**

CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to authenticate user information.

Two-way

**Three-way**

Four-way

Five-way

**Question No:22**

**(Marks:1)**

**Vu-Topper RM**

Possession of TOKEN gives the station the right to \_\_\_\_\_.

Modify the data

**Send the data**

Delete the data

Receive the data

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**Question No:23**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

Reservation

**Polling**

Token

None of the given

**Question No:24**

**(Marks:1)**

**Vu-Topper RM**

If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error locations.

**16**

2

4

8

**Question No:25**

**(Marks:1)**

**Vu-Topper RM**

Checksum is an error-detection technique that can be applied to a message of -----\_length.

Exactly 4

Exactly 8

Exactly 32

**Any**

**Question No:26**

**(Marks:1)**

**Vu-Topper RM**

To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the \_\_\_\_\_ layer.

Physical

**Network**

Presentation

Data link

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**Question No:27** (Marks:1)

**Vu-Topper RM**

CDMA stands for \_\_\_\_\_.

Carrier Division Multiple Access

Carrier Data Multiple Access

Code Data Multiple Access

**Code Division Multiple Access**

**Question No:28** (Marks:1)

**Vu-Topper RM**

In Block coding we divide our message into blocks of k bits called as

Code words

**Data words**

Passwords

Cross word

**Question No:29**

**(Marks:1)**

**Vu-Topper RM**

Stop-and-Wait protocol is used for \_\_\_\_\_

**Both flow and error control**

In \_\_, each station is forced to send a frame only at beginning of time slot.

**Slotted ALOHA**

**Question No:30**

**(Marks:1)**

**Vu-Topper RM**

IN MAC, when nodes use a multi-point or broadcast link, we need a \_----  
-----to coordinate access to the link.

**Multiple-access protocol**

**Question No:31**

**(Marks:1)**

**Vu-Topper RM**

In order to find hamming distance between two pair of words, \_\_\_\_\_  
logical operation is used.

**XOR**

**Question No:32**

**(Marks:1)**

**Vu-Topper RM**

In the context of Ethernet implementations maximum medium length of 10  
base-F

**2000**

**Question No:33**

**(Marks:1)**

**Vu-Topper RM**

Normally the value of KAMAX in pure aloha is

**15**

**Question No:34**

**(Marks:1)**

**Vu-Topper RM**

-----uses thick coaxial cable.

**10BASE5**

**Question No:35**

**(Marks:1)**

**Vu-Topper RM**

In Ethernet frame, both destination and sender addresses are of length

**6bytes**

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**Question No:36**

**(Marks:1)**

**Vu-Topper RM**

In \_ each band is reserved for a specific station and it belongs to the station all the time

**FDMA**

**Question No:37**

**(Marks:1)**

**Vu-Topper RM**

In Ethernet, the source address field in the MAC frame is the \_ address

**Previous station physical**

**Question No:38**

**(Marks:1)**

**Vu-Topper RM**

A personal computer or workstation on an Ethernet network must have \_\_\_\_------ card

**NIC**

**Question No:39**

**(Marks:1)**

**Vu-Topper RM**

For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the \_\_\_\_\_.

**Frame size**

**Question No:40**

**(Marks:1)**

**Vu-Topper RM**

In the context of Ethernet address, the source address is always a -- address

**Multicast**

**Question No:41**

**(Marks:1)**

**Vu-Topper RM**

In case of higher data rate, the number of impacted bits will be \_\_\_\_\_.

**Higher**

**Question No:42**

**(Marks:1)**

**Vu-Topper RM**

In each character protocol each character of frame is encoded using \_\_\_\_ coding system .

**Bit-oriented**

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**Question No:43**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_ collision are avoided by deferring transmission even if the channel is idle

**The interframe space**

**Question No:44**

**(Marks:1)**

**Vu-Topper RM**

In the \_\_\_\_\_ method, the stations in a network are organized in a logical ring.

**Token-passing**

**Question No:45**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_uses fiber-optic cable

**10base-F**

**Question No:46**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_ a station monitors the medium after it sends a frame to see if the transmission was successful.

**CSMA/CD**

**Question No:47**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ was used as the medium in 10 Base-T.

**Twisted-pair cabling**

**Question No:48**

**(Marks:1)**

**Vu-Topper RM**

Mac address is of \_\_\_\_\_

**48 bits**

**Question No:49**

**(Marks:1)**

**Vu-Topper RM**

PAP and CHAP are used by \_\_\_\_\_ as Authentication Protocols.

**PPP**

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**Question No:50**

**(Marks:1)**

**Vu-Topper RM**

SONET stands for

**Synchronous optical network**

**Question No:51**

**(Marks:1)**

**Vu-Topper RM**

The \_ method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately

**1 persistent method**

**Question No:52**

**(Marks:1)**

**Vu-Topper RM**

Fast Ethernet has a data rate of \_\_\_\_ Mbps.

**100**

**Question No:53**

**(Marks:1)**

**Vu-Topper RM**

Works with topologies in which one device is designated as a primary station and the other devices are secondary stations

**None of the given**

**Question No:54**

**(Marks:1)**

**Vu-Topper RM**

In hexadecimal notation of Ethernet address, each byte is separated by ----- symbol.

**Colon**

**Question No:55**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is an Authentication Protocol, which uses two-step process to authenticate user information.

**PAP**

**Question No:56**

**(Marks:1)**

**Vu-Topper RM**

In random access method, when two or more than two nodes transmit at the same time when

**All the nodes in collision survives**

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**Question No:57**

**(Marks:1)**

**Vu-Topper RM**

----- uses thin coaxial cable.

**10BASE2**

**Question No:58**

**(Marks:1)**

**Vu-Topper RM**

One of the most common protocols for point-to-point access is \_\_\_\_\_.

**PPP**

**Question No:59**

**(Marks:1)**

**Vu-Topper RM**

POTS stand for \_?

**Plain old telephone system**

**Question No:60**

**(Marks:1)**

**Vu-Topper RM**

POTS stand for \_?

**Plain old telephone system**

**Question No:61**

**(Marks:1)**

**Vu-Topper RM**

Data Link layer receives services from the \_\_\_\_\_ layer.

**Network layer**

**Question No:62**

**(Marks:1)**

**Vu-Topper RM**

In \_ protocol, the sender sends one frame at a time and wants for an acknowledgment before sending the next one.

**Stop-and-wait**

**Question No:63**

**(Marks:1)**

**Vu-Topper RM**

Data link layer is divided into \_\_\_\_\_ sub layers

**Two**

**Question No:64**

**(Marks:1)**

**Vu-Topper RM**

IN MAC, when nodes use a multi-point or broadcast link, we need a \_\_\_\_\_ to coordinate access to the link.

**Multiple-access protocol**

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**Question No:65**

**(Marks:1)**

**Vu-Topper RM**

The original Ethernet technology with the data rate of 10 MBps is called\_\_\_\_\_?

**Standard Ethernet**

**Question No:66**

**(Marks:1)**

**Vu-Topper RM**

If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise

**More**

**Question No:67**

**(Marks:1)**

**Vu-Topper RM**

For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the

**Collision size**

**Question No:68**

**(Marks:1)**

**Vu-Topper RM**

In a linear block code, the \_\_\_\_\_ of any two valid codewords creates another valid codeword.

14.

**XOR**

**Question No:69**

**(Marks:1)**

**Vu-Topper RM**

Which one of the following is not a Channelization Protocol?

**CSMA**

**Question No:70**

**(Marks:1)**

**Vu-Topper RM**

In order to find the location of two errors in an eight-bit stream, we have to see \_\_\_\_\_ different combinations.

**28**

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**Question No:71**

**(Marks:1)**

**Vu-Topper RM**

In the context of Ethernet address, the destination address is always a \_\_\_\_\_ address.

**Unicast, multicast, broadcast**

**Question No:72**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

**Gigabit Ethernet**

**Question No:73**

**(Marks:1)**

**Vu-Topper RM**

If the harming distance between sent and received code word is \_\_\_\_\_ then it shows that received data is corrupted.

**Is not equal to 0**

**Question No:74**

**(Marks:1)**

**Vu-Topper RM**

Ethernet address is of \_\_\_\_ bytes

**6**

**Question No:75**

**(Marks:1)**

**Vu-Topper RM**

What was the main drawback in the aloha random access method-----

**Redundancy in data**

**Question No:76**

**(Marks:1)**

**Vu-Topper RM**

In telephone, network POP stands for

**Point-of presence**

**Question No:77**

**(Marks:1)**

**Vu-Topper RM**

Transmission technology based on Ethernet provides a data rate of 1 billion bits per second.

**Standard Ethernet**

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**Question No:78**

**(Marks:1)**

**Vu-Topper RM**

In pure aloha the vulnerable time is \_\_\_\_ the frame transmission time

**Two times**

**Question No:79**

**(Marks:1)**

**Vu-Topper RM**

DSL stands for \_\_\_\_\_.

**Digital subscriber line**

**Question No:80**

**(Marks:1)**

**Vu-Topper RM**

The original Ethernet technology with the data rate of 10 Mbps is called

**Standard ethernet**

**Question No:81**

**(Marks:1)**

**Vu-Topper RM**

Which of the following is a correct controlled access method?

**All options**

**Question No:82**

**(Marks:1)**

**Vu-Topper RM**

When we represent the data in CDMA if a station is idle then it sends

**0**

**Question No:83**

**(Marks:1)**

**Vu-Topper RM**

In telephone, network POP stands for \_\_\_\_\_.

**Point of presence**

**Question No:84**

**(Marks:1)**

**Vu-Topper RM**

What was the main drawback in the aloha random access method?

**Redundancy data rate**

**Question No:85**

**(Marks:1)**

**Vu-Topper RM**

VCI address is \_\_\_\_ address in virtual circuit approach.

Private

Variable

**Global**

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**Question No:86**

**(Marks:1)**

**Vu-Topper RM**

Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching

True

**False**

**Question No:87**

**(Marks:1)**

**Vu-Topper RM**

Like 10 Base 5, 10 Base 2 is a \_\_\_\_\_ topology LAN.

Ring

Mesh

Star

**Bus**

**Question No:88**

**(Marks:1)**

**Vu-Topper RM**

In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.

Same as

Three times

**Two times**

Four times

**Question No:89**

**(Marks:1)**

**Vu-Topper RM**

The transmission medium that carries the message is referred to as the Send and receive device

**Communication channel**

Protocol

Gateways

**Question No:90**

**(Marks:1)**

**Vu-Topper RM**

Analog signals are referring to be \_\_\_\_\_

**Continuous**

Discrete

Periodic

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**Question No:91**

**(Marks:1)**

**Vu-Topper RM**

There are \_\_\_\_\_ types of serial transmission.

3

4

**2**

None of the given

**Question No:92**

**(Marks:1)**

**Vu-Topper RM**

There are three types of Hubs that exist. Which of the following options correctly describes these three?

Passive, dormant, special

Active , dormant , passive

Passive, Active, Turbo

**Passive, Active, Intelligent**

**Question No:93**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is the rate of change with respect to time.

Voltage

Time

Frequency

**Amplitude**

**Question No:94**

**(Marks:1)**

**Vu-Topper RM**

What is the frequency of a sine wave if it completes one cycle in 2 seconds?

**0.5**

0.6

0.3

0.2

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**Question No:95**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ relatively measures the strength of two signals.

Signal rate

Bit rate

Decibel

**Pulse Rate**

**Question No:96**

**(Marks:1)**

**Vu-Topper RM**

Central Hub in a Tree is an \_\_\_\_\_.

**Active hub**

passive hub

**Question No:97**

**(Marks:1)**

**Vu-Topper RM**

At the CRC generator, \_\_\_\_\_ added to the data unit before the divisional process.

**0s are**

1s are

a polynomial is

a CRC remainder is

**Question No:98**

**(Marks:1)**

**Vu-Topper RM**

A broadcast signal is received by the orbiting satellites which rebroadcasts \_\_\_\_\_.

to the resender

to the sender

to the atmosphere

**none of the given**

**Question No:99**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_, a signal is directed straight from antenna to antenna.

**Line of sight propagation**

Ground propagation

Space propagation

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**Question No:100**

**(Marks:1)**

**Vu-Topper RM**

Which of the following consist of just one redundant bit per data unit?

Two-dimensional Parity check.

CRC

**Simple Parity Check**

Checksum

**Question No:101**

**(Marks:1)**

**Vu-Topper RM**

The term \_\_\_\_\_ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

Packet Error

Burst Error

**Single Bit Error**

Character Error

**Question No:102**

**(Marks:1)**

**Vu-Topper RM**

Flow control is needed to prevent \_\_\_\_\_.

Bit errors

Overflow of the sender buffer

**Overflow of the receiver buffer**

Collision between sender and receiver

**Question No:103**

**(Marks:1)**

**Vu-Topper RM**

A fiber -optic cable transmit signals in the form of \_\_\_\_\_.

**Light**

Sound

Wave

None of the given

**Question No:104**

**(Marks:1)**

**Vu-Topper RM**

The inversion of the level at 1 bit is called as \_\_\_\_\_.

NRZ-L

**NRZ-I**

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**Question No:105**

**(Marks:1)**

**Vu-Topper RM**

What was the main draw back in ALOHA Random-Access method?

Redundancy in Data

Potential Collision

Inconsistent Data rate

**All of the given**

**Question No:106**

**(Marks:1)**

**Vu-Topper RM**

Data is transmitted using light through a \_\_\_\_\_ cable.

twisted pair

**fiber-optic**

coaxial

microwave

**Question No:107**

**(Marks:1)**

**Vu-Topper RM**

In OSI model headers are added at layer \_\_\_\_\_.

1,2,3 only

6,5,7 only

5,4,3 only

**6,5,4,3,2 only**

**Question No:108**

**(Marks:1)**

**Vu-Topper RM**

Is there any difference between Data communication and telecommunication are network?

Yes

**No**

**Question No:109**

**(Marks:1)**

**Vu-Topper RM**

In ASK, both \_\_\_\_\_ and \_\_\_\_\_ remain constant.

Amplitude, Frequency

Amplitude, phase

**Frequency, Phase**

Amplitude, Samples

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**Question No:110**

**(Marks:1)**

**Vu-Topper RM**

Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?

Checksum

Cyclic Redundancy Check (CRC)

**Longitudinal Red Check (LRC)**

Vertical Redundancy Check (VRC)

**Question No:111**

**(Marks:1)**

**Vu-Topper RM**

Baud rate is greater than or equal to the bit rate.

True

**False**

**Question No:112**

**(Marks:1)**

**Vu-Topper RM**

The most common type of connector used by coaxial cable is \_\_\_\_\_.

**BNC**

RJ-45

RJ-11

RJ-57

**Question No:113**

**(Marks:1)**

**Vu-Topper RM**

What is the period of a Sine wave having frequency of 10 Hz?

0.2

**0.1**

0.5

0.3

**Question No:114**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_, both peak amplitude and frequency remains constant as the phase changes.

**PSK**

QAM

Both PSK & QAM

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**Question No:115**

**(Marks:1)**

**Vu-Topper RM**

There are \_\_\_\_\_types of guided media.

2

**3**

4

5

**Question No:116**

**(Marks:1)**

**Vu-Topper RM**

TROPOSHERE is the layer of atmosphere above the troposphere but below space

True

**False**

**Question No:117**

**(Marks:1)**

**Vu-Topper RM**

Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as \_\_\_\_\_

DCE

**DTE**

DCE & DTE

None of the given

**Question No:118**

**(Marks:1)**

**Vu-Topper RM**

Normally the value of KAMAX in pure ALOHA is \_\_\_\_\_.

25

10

**15**

30

**Question No:119**

**(Marks:1)**

**Vu-Topper RM**

Sampling means measuring \_\_\_\_\_ of signal at equal intervals

**Amplitudes**

Frequencies

Phases

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**Question No:120**

**(Marks:1)**

**Vu-Topper RM**

A sine wave is \_\_\_\_\_.

periodic and discrete

aperiodic and discrete

**periodic and continuous**

aperiodic and continuous

**Question No:121**

**(Marks:1)**

**Vu-Topper RM**

To make sure that source does not overwhelm destination by sending data faster than it can be handled and processed is called?

Addressing & Routing

Exchange Management

**Flow Control**

Recovery

**Question No:122**

**(Marks:1)**

**Vu-Topper RM**

If a digital signal has "L" number of levels. \_\_\_\_\_ is the number of bits required to represent each level.

**Log L**

Log Base 2 L

L Square

Log(2\*L)

**Question No:123**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ category of coaxial cable is used for Cable TV.

RG-58

RG-59

**RG-11**

**Question No:124**

**(Marks:1)**

**Vu-Topper RM**

The level of signal is always positive in NRZ encoding.

True

**False**

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**Question No:125**

**(Marks:1)**

**Vu-Topper RM**

Radio is an example of \_\_\_\_\_ signal conversion.

**Analog to analog**

Analog to digital

Digital to Digital

Digital to analog

**Question No:126**

**(Marks:1)**

**Vu-Topper RM**

Analog To Analog Conversion Methods include \_\_\_\_\_.

**AM, PM and FM**

AM, FSK and PM

AM, PM and QAM

None of the given

**Question No:127**

**(Marks:1)**

**Vu-Topper RM**

The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?

**Standard Ethernet**

Fast Ethernet

Gigabit Ethernet

10 Gigabit Ethernet

**Question No:128**

**(Marks:1)**

**Vu-Topper RM**

Which of the following is not a characteristic of a sine wave?

Amplitude

**Segmentation**

Phase

Frequency

**Question No:129**

**(Marks:1)**

**Vu-Topper RM**

By using-----system, change in the file contents during file transmission can be avoided.

**Error Detection and Correction**

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:130**

**(Marks:1)**

**Vu-Topper RM**

In order to find the location of two errors in an eight bit stream, we have to see \_\_\_\_\_ different combinations.

**28**

16

8

4

**Question No:131**

**(Marks:1)**

**Vu-Topper RM**

In CSMA/CA, \_\_\_\_\_ is the amount of time divided into slots.

Interframe Window

**Contention Window**

Fragmented Window

Collided Window

**Question No:132**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

Frame sequencing

Acknowledgement numbering

Counter and Timer reset

**Frame sequencing and Acknowledgement numbering**

**Question No:133**

**(Marks:1)**

**Vu-Topper RM**

Which one of the following is a correct-access method?

Polling

Token Passing

**Reservation**

All of the given

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:134**

**(Marks:1)**

**Vu-Topper RM**

In the encoding where positive to negative transition is one and vice versa is zero.

Manchester

Differential Manchester

All of the given

**None of given**

**Question No:135**

**(Marks:1)**

**Vu-Topper RM**

If the window size is 63 in go-back N ARQ then what is the range of sequence number?

**0 to 63**

0 to 64

1 to 63

1 to 64

**Question No:136**

**(Marks:1)**

**Vu-Topper RM**

At the CRC checker, \_\_\_\_\_ means that the data unit is damaged.

A string of 0s

A string of 1s

A string of alternating 1s and 0s

**A nonzero remainder**

**Question No:137**

**(Marks:1)**

**Vu-Topper RM**

Human voice is example of \_\_\_\_\_ signal.

Digital

**Analog**

Discrete

None of the given

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:138**

**(Marks:1)**

**Vu-Topper RM**

In statistical time division multiplexing, number of slots in frame are less than \_\_\_\_\_.

Output data rate

**Input lines**

Input frequency

Multiplexed T lines

**Question No:139**

**(Marks:1)**

**Vu-Topper RM**

A \_\_\_\_\_ modem is used to connect two DTEs directly.

**'NULL**

Cable

DSL

None of the given

**Question No:140**

**(Marks:1)**

**Vu-Topper RM**

Asynchronous TDM is efficient only when the size of the time slot is kept relatively \_\_\_\_\_.

**Large**

Small

Medium

Zero

**Question No:141**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

$N+1$

$N-1$

**$N(N+1)$**

None of the given.

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:142**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is sometimes called the bit rate.

signal rate

modulation rate

**Data rate**

pulse rate

**Question No:143**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ category of coaxial cable is used for thin Ethernet.

**RG-58**

RG-59

RG-1

RG-47

**Question No:144**

**(Marks:1)**

**Vu-Topper RM**

Which one is not the function of data link layer?

**Line discipline**

Flow control

Error control

Network control

**Question No:145**

**(Marks:1)**

**Vu-Topper RM**

Session layer is responsible for \_\_\_\_\_.

Reassembly of data

**Maintaining the connection**

**Question No:146**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ mode of serial transmission guarantees fixed rate of data.

Synchronous

Asynchronous

**Isochronous**

Metachronous

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:147**

**(Marks:1)**

**Vu-Topper RM**

Data Link layer provides services to the \_\_\_\_\_ layer.

Application

Transport

**Network**

Session

**Question No:148**

**(Marks:1)**

**Vu-Topper RM**

The \_\_\_\_\_ is the number of signal elements sent per unit time.

Data rate

**Signal rate**

Bit rate 4. Period

**Question No:149**

**(Marks:1)**

**Vu-Topper RM**

The logical connection between the peer layers is \_\_\_\_\_ connection.

Physical

**Direct**

Indirect

Tangible

**Question No:150**

**(Marks:1)**

**Vu-Topper RM**

What is the period of a Sine wave having frequency of 5 Hz?

**0.2**

0.1

0.5

0.3

**Question No:151**

**(Marks:1)**

**Vu-Topper RM**

A sine wave is defined by \_\_\_\_\_ characteristics.

2

**3**

4

5

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:152**

**(Marks:1)**

**Vu-Topper RM**

If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise.

Constant

**More**

Less

Zero

**Question No:153**

**(Marks:1)**

**Vu-Topper RM**

The \_\_\_\_\_ generates the data and passes it along with any control information to a \_\_\_\_\_.

**DTE, DCE**

DCE, DTE

DTE, DTE

DCE, DCE

**Question No:154**

**(Marks:1)**

**Vu-Topper RM**

What is the period of a Sine wave having frequency of 2 Hz?

0.2

0.1

**0.5**

0.3

**Question No:155**

**(Marks:1)**

**Vu-Topper RM**

WDM stands for \_\_\_\_\_.

Wideband De-Modulation

Worst Data Manipulation

**Wavelength Division Multiplexing**

None of the given

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:156** (Marks:1)

**Vu-Topper RM**

Latency is made of \_\_\_\_\_ components.

**Two**

Three

Four

Five

**Question No:157** (Marks:1)

**Vu-Topper RM**

The logical connection between the peer layers is \_\_\_\_\_ connection.

Physical

Direct

Indirect

**Tangible**

**Question No:159** (Marks:1)

**Vu-Topper RM**

In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_ frame.

**EOT**

EKT

ENT

ESP

**Question No:160** (Marks:1)

**Vu-Topper RM**

\_\_\_\_\_ of a signal is the collection of all the component frequencies it contains.

Time domain

Frequency domain

**Frequency spectrum**

None of the given

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:161**

**(Marks:1)**

**Vu-Topper RM**

In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as \_\_\_\_\_.

Encapsulation

De-encapsulation

**Packetizing**

Framing

**Question No:162**

**(Marks:1)**

**Vu-Topper RM**

In circuit switched networks we have low efficiency but minimal \_\_\_\_

Delay

**Speed**

Throughput

Errors

**Question No:163**

**(Marks:1)**

**Vu-Topper RM**

AM signal requires \_\_\_\_\_ the BW of original signal

**Twice**

Half

Same

Three Times

**Question No:164**

**(Marks:1)**

**Vu-Topper RM**

Asynchronous transmission is \_\_\_\_\_.

**Slow**

Fast

Costly

None effective

**Question No:165**

**(Marks:1)**

**Vu-Topper RM**

The Internet is-----.

**A global network of computers networks**

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:166**

**(Marks:1)**

**Vu-Topper RM**

Time division multiplexing is used in \_\_\_\_\_ systems.

Analog

Hybrid

**Digital**

Automated

**Question No:167**

**(Marks:1)**

**Vu-Topper RM**

Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.

Two

Three

Four

**Five**

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**Question No:168**

**(Marks:1)**

**Vu-Topper RM**

At the Application layer, object/information is in the form of \_\_\_\_\_  
Message

**Packet**

Segment

Frame

**Question No:169**

**(Marks:1)**

**Vu-Topper RM**

POST stand for \_\_\_\_\_?

Plain Old Telephone Set

Plain Old Time Sharing

Plain Old Telephone System

**Plain Old Telephone Service**

**Question No:170**

**(Marks:1)**

**Vu-Topper RM**

Which of the following statement is correct?

**Digital signals are less prone to Noise**

Digital signals are highly prone to Noise

Analog Signals are less prone to Noise

Digital signals are not affected by Noise

**Question No:171**

**(Marks:1)**

**Vu-Topper RM**

In selective-reject ARQ, only the specific damaged or lost frame is--

**Retransmitted**

Forwarded

Selected

Rejected

**Question No:172**

**(Marks:1)**

**Vu-Topper RM**

In ASK correct formula for calculating the bandwidth is as \_\_\_\_\_.

**$B=(1+d)S$**

$B=(1*d)S$

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:173**

**(Marks:1)**

**Vu-Topper RM**

Encoding digital data into digital signals is called \_\_\_\_\_.

Analog-to-Digital Conversion

**Digital-to-Digital Conversion**

Digital-to-Analog Conversion

None of the given

**Question No:174**

**(Marks:1)**

**Vu-Topper RM**

Quadrature Amplitude Modulation (QAM) is the combination of \_\_\_\_\_ and \_\_\_\_\_.

**FSK, PSK**

PSK, FSK

ASK, PSK

ASK, FSK

**Question No:175**

**(Marks:1)**

**Vu-Topper RM**

PSK is susceptible to the noise degradation.

True

**False**

**Question No:176**

**(Marks:1)**

**Vu-Topper RM**

No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between \_\_\_\_\_.

Virtual LANs

**Two adjacent nodes**

Ethernet links

Source and destination PCs

**Question No:177**

**(Marks:1)**

**Vu-Topper RM**

GO-BACK-N protocol is one of the \_\_\_\_\_ layer protocols, which is used to control the flow of data during the transmission.

Physical

**Network**

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:178**

**(Marks:1)**

**Vu-Topper RM**

In \_\_\_\_\_ a dedicated communication path is established between two stations through the nodes of the network

**Circuit switching**

Packet switching

**Question No:179**

**(Marks:1)**

**Vu-Topper RM**

In frequency shift keying, \_\_\_\_\_ remain(s) constant.

Frequency

Amplitude

Both Amplitude and Phase

**Both Phase and Frequency**

**Question No:180**

**(Marks:1)**

**Vu-Topper RM**

We can have \_\_\_\_\_ different groups with a 4 bit block.

4

**8**

16

32

**Question No:181**

**(Marks:1)**

**Vu-Topper RM**

Line Configuration refers to the way two or more devices attach to a

Path

Circuit

**Link**

Router

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**Question No:182**

**(Marks:1)**

**Vu-Topper RM**

There are \_\_\_\_\_ basic categories of multiplexing.

2

**3**

7

5

**Question No:183**

**(Marks:1)**

**Vu-Topper RM**

Low pass channel has \_\_\_\_\_ bandwidth between two stations.

**Dedicated**

Shared

Multiplexed

Infinite

**Question No:184**

**(Marks:1)**

**Vu-Topper RM**

BSC supports \_\_\_\_\_ transmission using \_\_\_\_\_ flow control.

Full-duplex, Stop and wait ARQ

Half-duplex, Selective reject ARQ

Full-duplex, Selective reject ARQ

**Half-duplex, Stop and wait ARQ**

**Question No:185**

**(Marks:1)**

**Vu-Topper RM**

Quantization is a method of assigning \_\_\_\_\_ values in a specific range to sampled instances.

Negative

Positive

Binary

**Integral**

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**Question No:186**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ encoding is almost obsolete today

Bipolar

**Unipolar**

Polar

None of the given

**Question No:187**

**(Marks:1)**

**Vu-Topper RM**

YMODEM has \_\_\_\_\_ Byte of data unit.

**1024**

256

128

512

**Question No:188**

**(Marks:1)**

**Vu-Topper RM**

Like Data link layer, \_\_\_\_\_ is also responsible for Flow control.

**Transport Layer**

Session Layer

**Question No:189**

**(Marks:1)**

**Vu-Topper RM**

In CRC there is no error if the remainder at the receiver is

\_\_\_\_\_.

Equal to the remainder at the sender

**Zero**

Nonzero

The quotient at the sender

**Question No:190**

**(Marks:1)**

**Vu-Topper RM**

\_\_\_\_\_ is normally used where speed is priority in data transfer.

**Serial data transmission**

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**Question No:191**

**(Marks:1)**

**Vu-Topper RM**

A portion of the path that carries TX between a given pair of devices is known as \_\_\_\_\_.

Direct Point

Bridge

**Channel**

Access Point

**Question No:192**

**(Marks:1)**

**Vu-Topper RM**

What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?

100

500

300

**600**

**Question No:193**

**(Marks:1)**

**Vu-Topper RM**

Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of \_\_\_\_\_.

Local Area Network

**Metropolitan Area Network**

Wide Area Network

Home Based Network

**Question No:194**

**(Marks:1)**

**Vu-Topper RM**

Which layer is concerned with the syntax and semantics of info exchange between two systems?

Physical

Application

Session

**Presentation**

**For More Help Vu- Topper RM Contact What's app 03224021365**

**Question No:195**

**(Marks:1)**

**Vu-Topper RM**

Time domain shows changes in \_\_\_\_\_.

time w.r.t signal amplitude

**signal amplitude w.r.t time**

time

All of the given

**Question No:196**

**(Marks:1)**

**Vu-Topper RM**

In analog transmission, the base signal is called the\_\_\_\_\_.

**Carrier Signal**

Analog signal

Digital signal

Modulated signal

**Question No:197**

**(Marks:1)**

**Vu-Topper RM**

Which of the following is most affected by noise?

PSK

**ASK**

FSK

QAM

**Question No:198**

**(Marks:1)**

**Vu-Topper RM**

In Synchronous TDM, Time slots are not same in number as input devices.

True

**False**

**Question No:199**

**(Marks:1)**

**Vu-Topper RM**

Guard bands are used in \_\_\_\_\_ multiplexing technique, to avoid overlapping of frequency bands assigned to each user.

**FDM**

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**Question No:200**

**(Marks:1)**

**Vu-Topper RM**

DLC in Data Link Layer stands for \_\_\_\_\_.

Divide Line Communication

Data Line Code

**Data Link Control**

Demand Link Coordination

**Question No:201**

**(Marks:1)**

**Vu-Topper RM**

In Pulse Code Modulation (PCM), the sampling is dependent on---

Time

Amplitude

Frequency

**Signal Rate**

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**Question No:202**

**(Marks:1)**

**Vu-Topper RM**

Two or more computers connected so that they can communicate with each other and share information is called a \_\_\_\_\_.

satellite

protocol

broadcast

**network**

**Question No:203**

**(Marks:1)**

**Vu-Topper RM**

Analog refers to something that is continuous in \_\_\_\_\_.

**Time**

Space Frequency

None of the given

**Question No:204**

**(Marks:1)**

**Vu-Topper RM**

What is the frequency of a sine wave if it completes one cycle in 5 seconds?

0.5

0.6

0.3

**0.2**

**Question No:205**

**(Marks:1)**

**Vu-Topper RM**

In an analog hierarchy to carry voice channels, a group can carry \_\_\_\_\_ voice channels.

60

12

**20**

**Question No:206**

**(Marks:1)**

**Vu-Topper RM**

STP is more expensive than UTP and is less susceptible to noise.

**True**

False

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**Question No:207**

**(Marks:1)**

**Vu-Topper RM**

-----are used to exchange session management and control information between connected devices.

I-frames

**U-frames**

S-frames

N-frames

**Question No:208**

**(Marks:1)**

**Vu-Topper RM**

The message 1110 sent by a source is received by a destination as 1011.

This is \_\_\_\_\_ type of error.

**Burst**

Single-Bit

Uni-Bit

Hamming

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# CS601MCQs Final Term Updated Book By Junaid

Business Communication (Virtual University of Pakistan)

# AL-JUNAID INSTITUTE

## CS601 for FINAL TERM MCQs + Subjective (ACCORDING to UPDATED BOOK)

### Question No: 1

Ethernet Address is of-----bytes

- 4
- **6**
- 8
- 12

### Question No: 2

In Pure ALOHA, the vulnerable time is ----- the frame transmission time.

- Same as
- **Two times**
- Three times
- Four times

### Question No: 3

\_\_\_\_\_ was used as a medium in 10 Base-T

- Thick coaxial Cable
- Think coaxial Cable
- **Twisted Pair Cable**
- Two Mode For Cable

### Question No: 4

When data are transmitted from device A to device B, the header from A's layer 4 is read by B's \_\_\_\_\_ layer.

- **physical**
- transport
- application
- none of the given

### Question No: 5

In \_\_\_\_\_ encoding, we use three levels: positive, zero, and negative.

- polar
- nonpolar
- mmultilevel
- **bipolar**

### Question No: 6

A periodic signal completes one cycle in 0.001 s. What is the frequency?

- 1 Hz
- **100 Hz**
- 1 KHz
- 1 MHz

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## Question No: 7

There are \_\_\_\_\_ basic modes for the propagation of light in fiber-optic cables.

- 3
- 4
- 5
- **2**

## Question No: 8

A parabolic dish antenna is a(n) \_\_\_\_\_ antenna.

- Omnidirectional
- Bidirectional
- Horn
- **Unidirectional**

## Question No: 9

Zero crossing bandwidth is also called as equivalent noise bandwidth.

- True

- **False**

## Question No: 10

During the transmission, transmitting device produces a high frequency signal which acts as a basis for the transmission. This base signal is called \_\_\_\_\_.

- Analog signal
- **Digital signal**                      Not Confirm
- Modulated signal
- Carrier signal

## Question No 11:

There are \_\_\_\_\_ types of serial transmission:

- 1
- **2**
- 3
- 4

## Question No 12:

Synchronous transmission have \_\_\_\_\_.

- a start bit
- a stop bit
- gaps between bytes
- **none of the given**

## Question No 13

Using \_\_\_\_\_ encoding scheme in digital transmission, we represent 0 by zero voltage level and represent 1 by any positive voltage level.

- Polar
- AMI

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- Bipolar
- Unipolar

## Question No 14

\_\_\_\_\_ is the formula for finding number of bits sent per level.

- Log L
- Log2
- Log2\*2
- Log2 L

## Question No 15

Which of the following is an example of ITU-T modem standards:?

- T-series
- X-series
- N-series
- V-series

## Question No 16

The maximum data rate in the uploading direction is still \_\_\_\_\_.

- 26.6 Kbps
- 33.6 Kbps
- 36.6 Kbps
- 46.6 Kbps

## Question No 16

In \_\_\_\_\_ each station is forced to send a frame only at beginning of time slot

- Pure ALOHA
- Slotted ALOHA
- Fragmented ALOHA
- None of the given

## Question 17

In \_\_\_\_\_ each station sends a frame whenever it has a frame to send

- Pure ALOHA
- Slotted ALOHA
- Fragmented ALOHA
- None of the given

## Question No 18

The data rate of a T-1 line is \_\_\_\_\_.

- 1.544 Gbps
- 2.544 Mbps
- 1.544 Kbps
- 1.544 Mbps

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## Question 19

Circuit switched networks have \_\_\_\_\_ efficiency but \_\_\_\_\_ delay compared to the packet switched networks.

- high, minimal
- high, high
- low, minimal

## Question No 20

Category 5 UTP cable is used for data transmission of upto \_\_\_\_\_.

- 100 Mbps
- 200 Mbps
- 250 Mbps
- 400 Mbps

## Question No 21

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- Conical
- Barrel
- Circular
- Rectangular

## Question No 22

SONET stands for \_\_\_\_\_

- synchronous optical network
- synchronous operational network
- stream optical network
- shell operational network

## Question No 23

Fast Ethernet has a data rate of \_\_\_\_\_ Mbps.

- 10
- 100
- 1000
- 10,000

## Question No 24

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

- Ground
- Sky
- Line of sight
- Space

## Question No 25

In Bit-oriented framing, \_\_\_\_\_ bit pattern is used as a flag to define beginning and end of a frame.

- (11111111)<sub>2</sub>
- (10000001)<sub>2</sub>

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➤ Single

➤ (01111110)<sub>2</sub>

Question No 26

Traditionally, \_\_\_\_\_ protocols have been defined for the data-link layer to deal with flow and error control.

- Two
- Three
- Five
- Four

Question No 27

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across \_\_\_\_\_ data link

➤ Single

- Multi
- Single and Multi
- none of the given

Question No 28

Need for Addressing makes Asynchronous TDM inefficient for bit or byte \_\_\_\_\_.

- Interleaving
- Addition
- Subtraction
- None of the given

Question No 29

\_\_\_\_\_ takes data from one high speed line and breaks it into portions.

- ▶ Multiplexing
- ▶ Inverse multiplexing
- ▶ Inverse subtraction
- ▶ Inverse addition

Question No 30

For Carrier Sense Multiple Access/Collision Detection (CSMA/CD) we need a restriction on the \_\_\_\_\_

- Collision Size
- Signal Size
- Frame Size
- Station Size

Question No 31:

In Y-MODEM Multiple files can be sent simultaneously

- ▶ True
- ▶ False

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## Question No 32

At the data-link layer, \_\_\_\_\_ protocol is designed to handle both flow and error control, but communication is one frame at a time.

- Simple
- Go-Back-N
- Selective-Repeat
- **Stop-and-Wait**

## Question No 33

The PPP protocol uses \_\_\_\_\_, which is responsible for establishing, maintaining, configuring, and terminating links.

- Internet Protocol Control Protocol (IPCP)
- Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocols (NCPs)
- **Link Control Protocol (LCP)**

## Question No 34

Aloha, CSMA/CD and CSMA/CA are part of \_\_\_\_\_.

- Controlled Access protocol
- Channelization protocol
- Mutual access protocol
- Random Access protocol**

## Question No 35

Which of the following is an example of ITU-T modem standards:

- **T-series**
- X-series
- N-series
- V-series

## Question No 36

The maximum data rate in the uploading direction is still \_\_\_\_\_.

- 26.6 Kbps
- **33.6 Kbps**
- 36.6 Kbps
- 46.6 Kbps

## Question No 37

Category 5 UTP cable is used for data transmission of upto \_\_\_\_\_.

- **100 Mbps**
- 200 Mbps
- 250 Mbps
- 400 Mbps

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Question No 38

ALOHA the earliest random access method was developed in early\_\_\_\_\_.

1950s

1960s

1980s

1970s

Question No 39

In block coding, we divide our message into blocks, each of \_\_\_\_\_ bits; and add \_\_\_\_\_ redundant bits to each block to make the codewords of \_\_\_\_\_ bit length.

m, r, c

n, k, r

n, r, k

k, r, n

Question No 40

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

➤ Conical

➤ Barrel

➤ Circular

➤ Rectangular

Question No 41

MAC stand for \_\_\_\_\_

➤ Media Access Controller

➤ Main Access Control

➤ Mandatory Access Control

➤ Media Access Control

Question No 42

Which one of the following is a Random-Access protocol

➤ CDMA/CA

➤ ALOHA

➤ CDMA/CD

➤ TDMA

Question No 43

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

➤ Ground

➤ Sky

➤ Line of sight

➤ Space

Question No 44

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across \_\_\_\_\_ data link

➤ Single

➤ Multi

➤ Single and Multi

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- none of the given

Question No 45

Need for Addressing makes Asynchronous TDM inefficient for bit or byte \_\_\_\_\_.

- Interleaving
- **Addition**
- Subtraction
- None of the given

Question No 46

CDMA stands for \_\_\_\_\_

- Carrier Data Multiple Access
- **Code Division Multiple Access**
- Code Data Multiple Access
- Carrier Division Multiple Access

Question No 47

In Ethernet, the source address field in the MAC frame is the \_\_\_\_\_ address.

- Sender's logical
- **Previous station's physical**
- Next destination's physical
- Sender's service port

Question No 48

\_\_\_\_\_ takes data from one high speed line and breaks it into portions.

- **Multiplexing**
- Inverse multiplexing
- Inverse subtraction
- Inverse addition

Question No 49

A traditional telephone line has a bandwidth of \_\_\_\_\_.

- ▶ **2000Hz**
- ▶ 4000 Hz
- ▶ 2000 MHz
- ▶ 4000 MHz

Question No 50

To detect five errors, the Hamming distance between each pair of codewords should be at least\_\_\_\_\_.

- 4
- 5
- **11**
- 6

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## Question No 51

For FEC using XOR operation, we divide a packet into  $N$  chunks, and need to send \_\_\_\_\_ chunks to the receiver side.

- ☐  $N$
- ☐  $2N$
- ☐  $(N+1)/2$
- ☒  $N+1$

## Question No 52

If the ASCII character H is sent and the character I is received, what type of error is this?

- ☒ Single-bit
- ☐ Multiple-bit
- ☐ Burst
- ☐ Recoverable

## Question No 53

At the CRC generator, \_\_\_\_\_ added to the data unit after the division process

- ☐ 0s are
- ☐ 1s are
- ☐ The polynomial
- ☒ The CRC remainder is

## Question No 54

Error control in the data link layer is based on \_\_\_\_\_

- ☒ automatic repeat request
- ☐ automatic repeat acknowledgment
- ☐ automatic send acknowledgment

## Question No 55

In a technique called, compounding High-and-Low Resolution Packets, the low-resolution section in the first packet is \_\_\_\_\_.

- ☒ CRC Not Sure
- ☐ Parity code
- ☐ Empty

## Question No 56

In line discipline the initiator first transmits a frame called an \_\_\_\_\_

- ☐ Enquiry
- ☐ Acknowledgment
- ☒ NAK
- ☐ Reques

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**Question No 57**

Primary device uses \_\_\_\_\_ to receive transmission from the secondary devices.

- ACK Not Sure
- ENQ
- POLL

**Question No 58**

Token Bus has no commercial application in data communications

- True
- False

**Question No 59**

HDLC is an acronym for \_\_\_\_\_.

- High-duplex line communication
- High-level data link control
- Half-duplex digital link combination
- Host double-level circuit

**Question No 59**

The BNC-T connector is a T-shaped device with \_\_\_\_\_ ports

- Three
- Two
- Four
- Five

**Question No 60**

Each station in the Token Ring regenerates the frame.

- True
- False

**Question No 61**

Repeater works on \_\_\_\_\_ layer.

- Data Link
- Physical
- Network
- Application

**Question No 62**

Trunks are transmission media such as \_\_\_\_\_ that handle the telephone to the nearest end office.

- Satellite links
- Twisted-pair & Fiber-optic
- Twisted-pair
- Fiber-optic

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## Question No 63

At the data-link layer, \_\_\_\_\_ protocol is designed to handle both flow and error control, but communication is one frame at a time.

- Simple
- Go-Back-N
- Selective-Repeat
- **Stop-and-Wait**

## Question No 64

The PPP protocol uses \_\_\_\_\_, which is responsible for establishing, maintaining, configuring, and terminating links.

- Internet Protocol Control Protocol (IPCP)
- Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocols (NCPs)
- **Link Control Protocol (LCP)**

## Question No 65

The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?

- **Standard Ethernet**
- Fast Ethernet
- Gigabit Ethernet
- 10 Gigabit Ethernet

## Question No 66

Normally the value of Kmax in pure ALOHA is \_\_\_\_\_.

- 10
- **15**
- 25
- 30

## Question No 67

Which one of the following is not a Channelization Protocol?

- **CSMA**
- CDMA
- FDMA
- TDMA

## Question No 68

In CSMA/CA \_\_\_\_\_ is the amount of time divided into slots.

- **Contention Window**
- Interframe Window
- Fragmented Window
- Collided Window

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Question No 69

Aloha, CSMA/CD and CSMA/CA are part of \_\_\_\_\_.

- Controlled Access protocol
- Channelization protocol
- Mutual access protocol
- Random Access protocol**

Question No 70

ALOHA the earliest random access method was developed in early \_\_\_\_\_.

- 1950s
- 1960s
- 1980s
- 1970s**

Question No 71

For FEC using XOR operation, we divide a packet into N chunks, and need to send \_\_\_\_\_ chunks to the receiver side.

- N
- 2N
- $(N+1)/2$
- N+1**

Question No 71

The \_\_\_\_\_ layer is responsible for the creation and delivery of a frame to another node, along the link.

- Application
- Network
- Physical
- Data-link**

Question No 72

IP addresses are the identifiers at the \_\_\_\_\_ layer.

- Physical
- Data-link
- Transport
- Network**

Question No 73

Each host or each interface of a router is assigned \_\_\_\_\_ address.

- A multicast
- A broadcast
- Both multicast or broadcast
- A unicast**

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Question No 74

The ARP protocol is one of the auxiliary protocols defined in the \_\_\_\_\_ layer, which maps an IP address to a logical-link address.

- Physical
- Data-link
- Transport
- Network**

Question No 75

Following are the examples of connecting devices EXCEPT \_\_\_\_\_ .

- Link Layer Switches
- Hub
- Routers
- **Wire**

Question No 76

\_\_\_\_\_ eliminates collision and connect heterogeneous devices.

- Hub
- Repeater
- Bridge
- Switch**

Question No 77

\_\_\_\_\_ creates looping problem in learning switch.

- Unicast
- Multicast
- None of the Given
- **Broadcast**

Question No 78

The term FTTC stands for \_\_\_\_\_ .

Front Technology Terminal Connection

Free Technology Transmission Connection

Fiber Terminal Transmission Commission

**Fiber to the Curb**

Question No 79

Multiple-access Protocol is divided into \_\_\_\_\_.

- Two categories
- Four categories

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Five categories

**Three categories**

Question No 80

The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_ Ethernet.

Fast

Gigabit

10 Gigabit

**Standard**

Question No 81

CSMA/CD is not used in \_\_\_\_\_ Ethernet.

Standard

Fast

Gigabit

**10 Gigabit**

Question No 82

The amplitude of a digital signal depends upon the \_\_\_\_\_ to represent a bit.

Phase

Wavelength

Bandwidth

**Voltage**

Question No 83

In IEEE 802.11, a station with \_\_\_\_\_ mobility can move from one BSS to another, but the movement is confined inside one ESS.

None of the choices are correct

No-transition

**ESS-transition**

BSS-transition

Question No 84

In IEEE 802.11, communication between two stations in two different BSSs usually occurs via two \_\_\_\_\_.

BSSs

ESSs

None of the given

**Aps**

Question No 89

A property of a signal where it is being received by receiver after reflection from different intermediate objects (Building, iron polls, walls) is called \_\_\_\_\_

☒ **Attenuation**

☐ Interference

☐ Error

☐ Multipath Propagation

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Question No 90

\_\_\_\_\_ medium is used in wireless LANs for transportation of data.

- Copper pair cable
- Optical fiber cable
- Coaxial cable

**Air**

Question No 91

In context of control signaling network, what does STP stand for \_\_\_\_\_

- Shielded twisted pair
- Signaling transmission point
- Signal transportation procedure

**○ Signal transfer point**

Question No 92

POP stands for \_\_\_\_\_

- Presentation of points
- Points of points
- Points of packets

**○ Points of Presence**

Question No 93

What does LATA stand for \_\_\_\_\_

- Local Area Transmission Arena
- Local-Access Transit Access
- Local-Areas Transport Areas

**➤ Local-Access Transport Areas**

Question No 94

Unipolar encoding scheme is not supported over the medium having \_\_\_\_\_.

- The capability of handling DC component
- Low speed
- High speed
- No capability of handling DC component**

Question No 95

The amplitude of a digital signal depends upon the \_\_\_\_\_ to represent a bit.

- ▶ Phase
  - ▶ Wavelength
  - ▶ Bandwidth

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## ► Voltage

Question No 96

Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol

- Application
- Network
- Session
- **Transport**

Question No 97

In \_\_\_\_\_ each station is allotted a time slot during which it can send data

- NDMA
- CDMA
- **TDMA**
- FDMA

Question No 98

In \_\_\_\_\_ collisions are avoided by deferring transmission even if the channel is idle

- The Slotted Frame
- The Deployment Frame
- The Contention Window
- **The Interframe Space**

Question No 99

In the context of Ethernet Implementations, maximum medium length of 10 Base-F was \_\_\_\_\_ meters.

- 500
- 1000
- **2000**
- 150

Question No 100

Collisions in CSMA/CA are avoided through \_\_\_\_\_

- The Interframe Space
- The Contention Window
- Acknowledgments
- **All of the given**

## Important Questions

1. Define Three Types of Cast
2. SONET 04 Layers Function
3. Flow Diagram for the CSMA/CD
4. Aloha (This Topic is Most Important)
5. Compare and contrast CSMA/CD with CSMA/CA
6. What is the difference between a BSS and an ESS
7. What is the access method used by wireless LANs?
8. RJ Stands for?
9. Write Full Form of Channelization Protocol

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10. What is ARP how does it works
11. Compare a piconet and a scatternet
12. What kind of error is undetectable by the checksum?
13. What are properties of signals?
14. Whether in Asynchronous or Synchronous TDM, addressing is used?
15. What is the basic purpose of Router?
16. Difference between Hub, Switch and Router
17. HDLC Frame
18. Difference between dataword and codeword
19. SERVICES PROVIDED BY DALA LINK
20. Why we need a Null Modem?
21. Count LRC for the following bits?  
10011010 10100101 1101 0110
22. what are the categories of multiplexing?
23. What are the three purposes of control frames at data link layer?
24. Compare line discipline methods ENQ/ACK and Poll/ Select?
25. what is the difference between character oriented and bit oriented protocols? what is the difference between character oriented and bit oriented protocols?
26. Illustrate the two strategies used to avoid collision.
27. Variable Framing Techniques
28. Configurations & Transfer Modes in HDLC
29. What is CONTROLLED ACCESS and its types?
30. Stop and Wait Flow Control
31. If four computers are connected with wireless NIC in a LAN architecture like isolated LAN in wired network. What do we call such networks?
32. Normally a telephone network is divided into several levels of switching offices, mention their names.
33. If we want to be able to detect 4-bit errors, what should be the minimum Hamming distance? Answer with the help of formula.
34. Distinguish between Slotted aloha and Pure aloha.
35. Which form of signal represents binary 1 and 0 by positive and negative voltage level?
36. Vulnerable time for pure ALOHA protocol
37. Space/time model of the collision in CSMA
38. CONTROLLED ACCESS
39. What is Token Passing
40. Synchronous and Asynchronous
41. What is OSI Layer?
42. Multiplexing
43. Switched Ethernet
44. Define the type of the following destination addresses:  
A. 4A: 30:10:21: 10:1A  
B. 47:20:1B:2E:08:EE

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C. FF: FF: FF: FF: FF: FF

45. Abbreviation of the following

- |        |                               |
|--------|-------------------------------|
| ➤ DLC  | (Data Link Control)           |
| ➤ FSM  | (Finite State Machine)        |
| ➤ POTS | (Plain old telephone service) |
| ➤ FTTC | (Fiber to the Curb)           |

**CS601 Finalterm Mcqs**

**Updated handouts 2022**

**#Final-Term**

**ORANGE MONKEY TEAM**

**Important NOTE:-**

Maybe toppers are successful in school or college carrier but backbenchers are successful in their life because toppers run according to books while backbenchers don't follow rules and regulation that is why backbenchers are successful. For life, there are no books and another reason why toppers can't succeed because they fear failure and don't try anything new. Backbenchers don't fear failure as they have faced the failure in their school or college carrier so that is the reason they try new things and at last gain success.

As Abdul Kalam nicely quoted, "The best brains of the nation may be found on the last benches of the classroom"

**Translation into Urdu:**

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**ORANGE MONKEY TEAM**

ہوسکتا ہے کہ ٹاپرز اسکول یا کالج کیریئر میں کامیاب ہوں لیکن بیک بینچر اپنی زندگی میں کامیاب ہوتے ہیں کیونکہ ٹاپر کتابوں کے مطابق چلتے ہیں جب کہ بیک بینچر قواعد و ضوابط پر عمل نہیں کرتے اسی وجہ سے بیک بینچر کامیاب ہوتے ہیں۔ زندگی کے لیے، کوئی کتابیں نہیں ہیں اور ایک اور وجہ ہے کہ ٹاپرز کامیاب نہیں ہو پاتے کیونکہ وہ ناکامی سے ڈرتے ہیں اور کچھ نیا کرنے کی کوشش نہیں کرتے۔ بیک بینچر ناکامی سے نہیں ڈرتے کیونکہ انہوں نے اپنے اسکول یا کالج کیریئر میں ناکامی کا سامنا کیا ہے یہی وجہ ہے کہ وہ نئی چیزیں آزما رہے ہیں اور آخر کار کامیابی حاصل کرتے ہیں۔

جیسا کہ عبدالکلام نے اچھی طرح سے نقل کیا ہے، "قوم کے بہترین دماغ کلاس روم کے آخری بینچوں پر مل سکتے ہیں۔"





1. In \_\_\_\_\_ each station is allocated a time slot during which it can send data  
TDMA.....confirm
2. PAP and CHAP are used by \_\_\_\_\_ as Authentication Protocols.  
PPP.....confirm
3. In each character protocol each character of frame is encoded using \_\_\_\_\_ coding system .  
Bit-oriented
4. IN MAC, when nodes use a multi-point or broadcast link, we need a \_\_\_\_\_ to coordinate access to the link.  
Multiple-access protocol/.....confirm
5. The original Ethernet technology with the data rate of 10 MBps is called \_\_\_\_\_?  
Standard Ethernet....confirm
6. \_\_\_\_\_ uses fiber-optic cable

**10base-F....confirm**

7. One of the most common protocols for point-to-point access is \_\_\_\_\_.

**PPP.....confirm**

8. \_\_\_\_\_ is an Authentication Protocol, which uses two-step process to authenticate user information.

**PAP.....confirm**

9. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise

**More.....confirm**

10. For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the

**Collision size....confirm**

11. Stop-and-Wait protocol is used for \_\_\_\_\_.

**Both flow and error control.....confirm**

12. In hexadecimal notation of Ethernet address, each byte is separated by \_\_\_\_\_ symbol.

**Colon....confirm**

13. In a linear block code, the \_\_\_\_\_ of any two valid codewords creates another valid codeword.

- 14.

**XOR.....confirm**

15. In the \_\_\_\_\_ method, the stations in a network are organized in a logical ring.

**Token-passing....confirm**

16. Which one of the following is not a Channelization Protocol?

**CSMA.....confirm**

17. In order to find the location of two errors in an eight bit stream, we have to see \_\_\_\_\_ different combinations.

**28....confirm**

18. If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error locations

**16.....confirm**

19. In \_\_\_\_\_ each station sends a frame whenever it has a frame to send.

**Slotted ALoHA.....confirm**

20. \_\_\_\_\_ is the process of converting digital data to digital signals.

**Line coding.....confirm**

21. In case of higher data rate, the number of impacted bits will be \_\_\_\_\_.

**Higher**

22. In \_ protocol, the sender sends one frame at a time and wants for an acknowledgment before sending the next one.

**Stop-and-wait**

23. In block coding we divide our message into blocks of K bits called as \_\_\_\_\_.

**Data words....confirm**

24. CDMA stands for \_\_\_\_\_.

**Code division Multiple Access.....confirm**

25. \_\_\_\_\_ is not a function of data link control.

**Modulation.....confirm**

26. CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to authenticate user information.

**Three-way.....confirm from net**

27. In the context of Ethernet address, the source address is always a \_\_\_\_\_ address.

Unicast...confirm

28. In the context of Ethernet address, the destination address is always a \_\_\_\_\_ address.

Unicast, multicast, broadcast....confirm

29. \_\_\_\_\_ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

Gigabit Ethernet....confirm

30. Fast Ethernet has a data rate of \_\_\_\_\_ Mbps.

100.....confirm

31. In random access method, when two or more than two nodes transmit at the same time when

All the nodes in collision survives

32. Data link layer is divided into \_\_\_\_\_ sub layers

Two.....confirm

33. If the hamming distance between sent and received code word is \_\_\_\_\_ then it shows that received data is corrupted.

(Is not equal to)0....confirm

34. In order to find hamming distance between two pair of words, \_\_\_\_\_ logical operation is used.

XOR....confirm

35. Ethernet address is of \_\_\_\_\_ bytes

6....confirm

36. In \_\_\_\_\_ a station monitors the medium after it sends a frame to see if the transmission was successful.

CSMA/CD.....confirm

37. For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the \_\_\_\_\_.

Frame size...confirm

38. Ethernet address is of \_\_\_\_\_ bytes

6

39. What was the main drawback in the aloha random access method

Redundancy in data

40. In telephone, network POP stands for

Point-of presence

41. Possession of token gives the station the right to \_\_\_\_\_.

Send the data

42. In Ethernet, the source address field in the MAC frame is the \_ address

Previous station physical

43. Normally the value of Kmax in pure aloha is

15

44. \_\_\_\_\_ uses thin coaxial cable.

10 base 5

45. MAC stands for \_\_\_\_\_

Media access control

46. In the context of Ethernet implementations maximum medium length of 10 base-F

2000

47. Transmission technology based on Ethernet provides a data rate of 1 billion bits per second.

### Standard Ethernet

48. Works with topologies in which one device is designated as a primary station and the other devices are secondary stations

None of the given

49. MAC address is of \_\_\_\_\_.

48 bits

50. The type of aloha which improves the efficiency of pure aloha is.

Slotted aloha

51. A personal computer or workstation on an Ethernet network must have \_\_\_\_\_ card

NIC

52. In which controlled access method all data exchange must be made through primary device even when the ultimate destination is a secondary device

Polling

53. In \_\_\_\_\_ each band is reserved for a specific station and it belongs to the station all the time

FDMA

54. In pure aloha the vulnerable time is \_\_\_\_\_ the frame transmission time

Two times

55. DSL stands for \_\_\_\_\_.

Digital subscriber line

56. In the context of Ethernet address, the source address is always a \_\_\_\_\_ address

Multicast

57. The original Ethernet technology with the data rate of 10 Mbps is called

Standard ethernet

58. Which of the following is a correct controlled access method

**All options**

**59. When we represent the data in CDMA if a station is idle then it sends**

**0**

**60. In telephone, network POP stands for \_\_\_\_\_.**

**Point of presence**

**61. What was the main drawback in the aloha random access method**

**Redundancy data rate**

**62. \_\_\_\_\_ transmission technology based on Ethernet provides a data rate of 1 billion bits per second**

**Gigabit Ethernet**

**63. In which controlled access method all data exchanges must be made through a primary device even when the ultimate destination is a secondary device.**

**Polling**

**64. In telephone, network POP stands for**

**Point of presence**

**65. Mac address is of \_\_\_\_\_**

**48 bits**

**66. In \_\_\_\_\_ collision are avoided by deferring transmission even if the channel is idle**

**The interframe space**

**67. POTS stand for \_?**

**Plain old telephone system.....confirm**

**68. Normally the value of Kmax in pure aloha is**

**15**

69. In Ethernet frame both destination and sender addresses are of length  
6 byte

70. In \_\_\_\_\_ each band is reserved for a specific station and it belongs to the station all the time

FDMA

71. The original Ethernet technology with a data rate of 10MBps is called

Standard Ethernet

72. The \_\_\_\_\_ method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately

1 persistent method

73. IN allocated a time slot during which it can send data each station is

TDMA

74. In which controlled access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device

Polling

75. CDMA stands for

Code data multiply access

76. Works with topologies in which one device is designated as a primary station and the other device are secondary stations

Polling

77. In Ethernet, the source address field in the MAC frame is the address

Next destination physical

1. VCI address is \_\_\_\_\_ address in virtual circuit approach.

1. Private

2. Variable

3. Local

4. Global

**2. Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching**

1. true

2. false

**3. In the \_\_\_\_\_ method, the stations in a network are organized in a logical ring.**

1. Token-passing

2. Polling

3. Reservation

4. None of the given

**4. Like 10 Base 5, 10 Base 2 is a \_\_\_\_\_ topology LAN.**

1. Ring

2. Mesh

3. Star

4. Bus

**5. In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.**

1. Same as

2. Three times

**3. Two times**

4. Four times

**6. The transmission medium that carries the message is referred to as the \_\_\_\_\_.**

1. send and receive device

**2. communication channel**

3. protocol

4. gateways

**7. Analog signals are refers to be \_\_\_\_\_**

**1. Continuous**

2. Discrete

3. Periodic

4. Aperiodic

**8. There are \_\_\_\_\_ types of serial transmission.**

1. 3

2. 4

**3. 2**

4. None of the given

9. To calculate the data rate for noisy channel \_\_\_\_\_ formula is used.

1. Shannon

2. Nyquist

3. Propagation

4. Greedy

10. There are three types of Hubs that exist. Which of the following options correctly describes these three?

1. Passive, dormant, special

2. Active , dormant , passive

3. Passive, Active, Turbo

4. Passive, Active, Intelligent

11. \_\_\_\_\_ is the rate of change with respect to time.

1. Voltage

2. Time

3. Frequency

4. Amplitude

12. To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the \_\_\_\_\_ layer.

1. Physical

2. Network

3. Presentation

4. Data link

**13. What is the frequency of a sine wave if it completes one cycle in 2 seconds?**

1. 0.5

2. 0.6

3. 0.3

4. 0.2

**14. \_\_\_\_\_ relatively measures the strength of two signals.**

1. Signal rate

2. Bit rate

3. Decibel

4. Pulse Rate

**15. Central Hub in a Tree is an \_\_\_\_\_.**

1. Active hub

2. passive hub

**16. At the CRC generator, \_\_\_\_\_ added to the data unit before the divisional process.**

1. 0s are

2. 1s are

3. a polynomial is

4. a CRC remainder is

17. A broadcast signal is received by the orbiting satellites which rebroadcasts \_\_\_\_\_.

1. to the resender
2. to the sender
3. to the atmosphere
4. none of the given

18. The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

1. Duplicated
2. Blocked
3. Cleaned
4. Escaped

19. In \_\_\_\_\_, a signal is directed straight from antenna to antenna.

1. Line of sight propagation
2. Ground propagation
3. Space propagation
4. Tropospheric propagation

20. Which of the following consist of just one redundant bit per data unit?

1. Two dimensional Parity check.

2. CRC

**3. Simple Parity Check**

4. Checksum

**21. The term \_\_\_\_\_ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.**

1. Packet Error

2. Burst Error

**3. Single Bit Error**

4. Character Error

**22. Flow control is needed to prevent \_\_\_\_\_.**

1. Bit errors

2. Overflow of the sender buffer

**3. Overflow of the receiver buffer**

4. Collision between sender and receiver

**23. Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol.**

1. Transport

2. Application

3. Session

**4. Network**

24. A fiber -optic cable transmit signals in the form of \_\_\_\_\_.

1. Light

2. Sound

3. Wave

4. None of the given

25. Possession of TOKEN gives the station the right to \_\_\_\_\_.

1. Modify the data

2. Send the data

3. Delete the data

4. Receive the data

26. The inversion of the level at 1 bit is called as \_\_\_\_\_.

1. NRZ-L

2. NRZ-I

3. RZ

4. None of the given.

27. What was the main draw back in ALOHA Random-Access method?

1. Redundancy in Data

2. Potential Collision

3. Inconsistent Data rate

4. All of the given

28. Data is transmitted using light through a \_\_\_\_\_ cable.

1. twisted pair

2. fiber-optic

3. coaxial

4. microwave

29. In \_\_\_\_\_, each station sends a frame whenever it has a frame to send.

1. Slotted ALOHA

2. Fragmented ALOHA

3. Pure ALOHA

4. None of the given

30. In OSI model headers are added at layer \_\_\_\_\_.

1. 1,2,3 only

2. 6,5,7 only

3. 5,4,3 only

4. 6,5,4,3,2 only

31. Is there any difference between Data communication and telecommunication are network?

1. Yes

2. No

32. \_\_\_\_\_ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

1. CHAP

**2. PAP**

3. LCP

4. ICMP

**33. Which level of the TCP/IP reference model routes data/information across a network channel?**

1. Application Layer

2. Data Link Layer

3. Transport Layer

**4. Network Layer**

**34. In ASK, both \_\_\_\_\_ and \_\_\_\_\_ remain constant.**

1. Amplitude, Frequency

2. Amplitude, phase

**3. Frequency, Phase**

4. Amplitude, Samples

**35. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?**

1. Checksum

2. Cyclic Redundancy Check (CRC)

**3. Longitudinal Red Check (LRC)**

4. Vertical Redundancy Check (VRC)

36. Baud rate is greater than or equal to the bit rate.

1. True

2. False

37. The most common type of connector used by coaxial cable is \_\_\_\_\_.

1. BNC

2. RJ-45

3. RJ-11

4. RJ-57

38. In even parity check code, the value of syndrome is \_\_\_\_\_ if the number of 1s is even.

1. 1

2. 2

3. 0

4. 4

39. What is the period of a Sine wave having frequency of 10 Hz?

1. 0.2

2. 0.1

3. 0.5

4. 0.3

40. In \_\_\_\_\_, both peak amplitude and frequency remains constant as the phase changes.

1. PSK

2. QAM

3. Both PSK & QAM

4. None of the given

41. There are \_\_\_\_\_ types of guided media.

1. 2

2. 3

3. 4

4. 5

42. \_\_\_\_\_ is responsible for governing node to node communication.

1. Application Layer

2. Session Layer

3. Data Link layer

4. Presentation Layer

43. TROPOSHERE is the layer of atmosphere above the troposphere but below space

1. True

2. False

44. Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as \_\_\_\_\_

1. DCE

2. DTE

3. DCE & DTE

4. None of the given

45. Normally the value of  $K_{max}$  in pure ALOHA is \_\_\_\_\_.

1. 25

2. 10

3. 15

4. 30

46. Sampling means measuring \_\_\_\_\_ of signal at equal intervals

1. Amplitudes

2. Frequencies

3. Phases

4. None of the given

47. A sine wave is \_\_\_\_\_.

1. periodic and discrete

2. aperiodic and discrete

3. periodic and continuous

4. aperiodic and continuous

48. The inversion of the level represents a 1 bit in \_\_\_\_\_ encoding

1. NRZ-I

2. NRZ-L

3. RZ

4. Manchester

49. To make sure that source does not overwhelm destination by sending data faster than it can be handled and processed is called?

1. Addressing & Routing

2. Exchange Management

3. Flow Control

4. Recovery

50. Time domain plot show changes in signal phase with respect to time.

1. true

2. false

51. In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

1. Polling

2. Reservation

3. Token Passing

4. Port Forwarding

52. If a digital signal has "L" number of levels. \_\_\_\_\_ is the number of bits required to represent each level.

1. Log L
2. Log Base 2 L
3. L Square
4. Log(2\*L)

53. \_\_\_\_\_ category of coaxial cable is used for Cable TV.

1. RG-58
2. RG-59
3. RG-11
4. RG-47

54. The level of signal is always positive in NRZ encoding.

1. true
2. false

55. Radio is an example of \_\_\_\_\_ signal conversion.

1. Analog to analog
2. Analog to digital
3. Digital to Digital
4. Digital to analog

56. Analog To Analog Conversion Methods include \_\_\_\_\_.

1. AM, PM and FM

2. AM, FSK and PM

3. AM, PM and QAM

4. None of the given

**57. The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?**

**1. Standard Ethernet**

2. Fast Ethernet

3. Gigabit Ethernet

4. 10 Gigabit Ethernet

**58. Which of the following is not a characteristic of a sine wave?**

1. Amplitude

**2. Segmentation**

3. Phase

4. Frequency

**59. By using \_\_\_\_\_ system, change in the file contents during file transmission can be avoided.**

1. Routing

2. Flow Control

**3. Error Detection and Correction**

4. Congestion Control

**60. In order to find the location of two errors in an eight bit stream, we have to see \_\_\_\_\_ different combinations.**

1. 28

2. 16

3. 8

4. 4

61. In CSMA/CA, \_\_\_\_\_ is the amount of time divided into slots.

1. Interframe Window

2. Contention Window

3. Fragmented Window

4. Collided Window

62. \_\_\_\_\_ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

1. Frame sequencing

2. Acknowledgement numbering

3. Counter and Timer reset

4. Frame sequencing and Acknowledgement numbering

63. Which one of the following is a correct-access method?

1. Polling

2. Token Passing

3. Reservation

4. All of the given

**64. In the encoding where positive to negative transition is one and vice versa is zero.**

1. Manchester
2. Differential Manchester
3. All of the given
4. None of given

**65. If the window size is 63 in go-back N ARQ then what is the range of sequence number?**

1. 0 to 63
2. 0 to 64
3. 1 to 63
4. 1 to 64

**66. Following characteristics fully describe \_\_\_\_\_,**  
**Amplitude**  
**Period/Frequency**  
**Phase**

1. Sine Waves
2. Digital Signals
3. Aperiodic Signals
4. None of the given

**67. Digitizing an Analog Signal is called \_\_\_\_\_.**

1. Analog-to-Digital Conversion
2. Digital-to-Digital Conversion

3. Digital-to-Analog Conversion

4. None of the given

**68. At the CRC checker, \_\_\_\_\_ means that the data unit is damaged.**

1. A string of 0s

2. A string of 1s

3. A string of alternating 1s and 0s

**4. A nonzero remainder**

**69. Which one of the following is a Random-Access protocol?**

1. TDMA

**2. CDMA/CD**

3. ALOHA

4. CDMA/CA

**70. Human voice is example of \_\_\_\_\_ signal.**

1. Digital

**2. Analog**

3. Discrete

4. None of the given

**71. In statistical time division multiplexing, number of slots in frame are less than \_\_\_\_\_.**

1. Output data rate

2. Input lines

3. Input frequency

4. Multiplexed T lines

72. A \_\_\_\_\_ modem is used to connect two DTEs directly.

1. 'NULL

2. Cable

3. DSL

4. None of the given

73. If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error locations.

1. 16

2. 2

3. 4

4. 8

74. Asynchronous TDM is efficient only when the size of the time slot is kept relatively \_\_\_\_\_.

1. Large

2. Small

3. Medium

4. Zero

75. The \_\_\_\_\_ protocol uses both flow and error control.

1. TCP/IP

2. HDLC

3. ICMP

**4. Stop-and-Wait**

76. In \_\_\_\_\_, each station is allocated a time slot during which it can send data.

1. CDMA

2. NDMA

**3. TDMA**

4. FDMA

77. The extra bits added with the original data for error detection/correction are called \_\_\_\_\_.

1. Fault bits

2. Port numbers

**3. Redundant bits**

4. Process IDs

78. \_\_\_\_\_ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

1.  $N+1$

2.  $N-1$

**3.  $N(N+1)$**

4.  $N \times N$

79. \_\_\_\_\_ is sometimes called the bit rate.

1. signal rate
2. modulation rate
3. Data rate
4. pulse rate

80. Collisions in CSMA/CA are avoided through \_\_\_\_\_.

1. The Interframe Space
2. The Contention Window
3. Acknowledgements
4. All of the given

81. \_\_\_\_\_ category of coaxial cable is used for thin Ethernet.

1. RG-58
2. RG-59
3. RG-1
4. RG-47

82. Which one is not the function of data link layer?

1. Line discipline
2. Flow control
3. Error control
4. Network control

83. Session layer is responsible for \_\_\_\_\_.

1. Reassembly of data
2. Maintaining the connection

84. \_\_\_\_\_ mode of serial transmission guarantees fixed rate of data.

1. Synchronous
2. Asynchronous
3. Isochronous
4. Metasochronous

85. Data Link layer provides services to the \_\_\_\_\_ layer.

1. Application
2. Transport
3. Network
4. Session

86. The \_\_\_\_\_ is the number of signal elements sent per unit time.

1. Data rate
2. Signal rate
3. Bit rate
4. Period

87. The logical connection between the peer layers is \_\_\_\_\_ connection.

1. Physical

**2. Direct**

3. Indirect

4. Tangible

**88. What is the period of a Sine wave having frequency of 5 Hz?**

**1. 0.2**

2. 0.1

3. 0.5

4. 0.3

**89. A sine wave is defined by \_\_\_\_\_ characteristics.**

1. 2

**2. 3**

3. 4

4. 5

**90. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise.**

1. Constant

**2. More**

3. Less

4. Zero

91. S-frames in High-Level Data Link Control (HDLC) are only used to transfer \_\_\_\_\_ information.

1. User data
2. Control
3. Redundant
4. Original

92. The \_\_\_\_\_ generates the data and passes it along with any control information to a \_\_\_\_\_.

1. DTE, DCE
2. DCE, DTE
3. DTE, DTE
4. DCE, DCE

93. The type of ALOHA which improves the efficiency of Pure ALOHA is \_\_\_\_\_.

1. Slotted ALOHA
2. Upper ALOHA
3. Integrated ALOHA
4. Lower ALOHA

94. What is the period of a Sine wave having frequency of 2 Hz?

1. 0.2
2. 0.1
3. 0.5

4. 0.3

95. WDM stands for \_\_\_\_\_.

1. Wideband De-Modulation

2. Worst Data Manipulation

3. Wavelength Division Multiplexing

4. None of the given

96. Latency is made of \_\_\_\_\_ components.

1. Two

2. Three

3. Four

4. Five

97. The logical connection between the peer layers is \_\_\_\_\_ connection.

1. Physical

2. Direct

3. Indirect

4. Tengible

98. In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_ frame.

1. EOT

2. EKT

3. ENT

4. ESP

99. \_\_\_\_\_ of a signal is the collection of all the component frequencies it contains.

1. Time domain

2. Frequency domain

3. Frequency spectrum

4. None of the given

100. CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to authenticate user information.

1. Two-way

2. Three-way

3. Four-way

4. Five-way

101. RZ stands for \_\_\_\_\_.

1. Return to Z

2. Retire a Zero

3. Return of Zero

4. None of the given

1. In \_\_\_\_\_ each band is reserved for a specific station, and it belongs to the station all the time.

1. FDMA

2. CDMA

3. TDMA

4. CSMA

**2. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as \_\_\_\_\_.**

1. Encapsulation

2. De-encapsulation

**3. Packetizing**

4. Framing

**3. In Block coding we divide our message into blocks of k bits called as \_\_\_\_\_.**

1. Code words

**2. Data words**

3. Passwords

4. Cross words

**4. In circuit switched networks we have low efficiency but minimal \_\_\_\_\_.**

1. Delay

**2. Speed**

3. Throughput

4. Errors

**5. Error correction is more \_\_\_\_\_ than the error detection.**

1. Easy
2. Useless
3. Informal
- 4. Difficult**

**6. AM signal requires \_\_\_\_\_ the BW of original signal**

- 1. Twice**
2. Half
3. Same
4. Three Times

**7. Asynchronous transmission is \_\_\_\_\_.**

- 1. Slow**
2. Fast
3. Costly
4. None effective

**8. The Internet is \_\_\_\_\_.**

1. software for sending e-mail around the world
2. a government-owned agency that links computers
- 3. a global network of computers networks**
4. a specialised form of local area network

**9. Time division multiplexing is used in \_\_\_\_\_ systems.**

1. Analog

2. Hybrid

**3. Digital**

4. Automated

**10. Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.**

1. Two

2. Three

3. Four

**4. Five**

**11. Checksum is an error-detection technique that can be applied to a message of \_\_\_\_\_ length.**

1. Exactly 4

2. Exactly 8

3. Exactly 32

**4. Any**

**12. At the Application layer, object/information is in the form of \_\_\_\_\_.**

1. Message

**2. Packet**

3. Segment

4. Frame

**13. POST stand for \_\_\_\_\_?**

1. Plain Old Telephone Set
2. Plain Old Time Sharing
3. Plain Old Telephone System
- 4. Plain Old Telephone Service**

**14. Which of the following statement is correct?**

- 1. Digital signals are less prone to Noise**
2. Digital signals are highly prone to Noise
3. Analog Signals are less prone to Noise
4. Digital signals are not affected by Noise

**15. In selective-reject ARQ, only the specific damaged or lost frame is \_\_\_\_\_.**

- 1. Retransmitted**
2. Forwarded
3. Selected
4. Rejected

**16. In ASK correct formula for calculating the bandwidth is as \_\_\_\_\_.**

- 1.  $B=(1+d)S$**
2.  $B=(1*d)S$
3.  $B=(d-1)S$

4.  $B=(d-5)S$

17. Encoding digital data into digital signals is called \_\_\_\_\_.

1. Analog-to-Digital Conversion

2. Digital-to-Digital Conversion

3. Digital-to-Analog Conversion

4. None of the given

18. If the Hamming distance between sent and received code word is \_\_\_\_\_, then it shows that received data is corrupted.

1.  $\neq 0$

2.  $= 0$

3.  $\neq 1$

4.  $= 1$

19. Quadrature Amplitude Modulation (QAM) is the combination of \_\_\_\_\_ and \_\_\_\_\_.

1. FSK, PSK

2. PSK, FSK

3. ASK, PSK

4. ASK, FSK

20. PSK is susceptible to the noise degradation.

1. True

2. False

21. \_\_\_\_\_ works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

1. Reservation
2. Polling
3. Token
4. None of the given

22. No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between \_\_\_\_\_.

1. Virtual LANs
2. Two adjacent nodes
3. Ethernet links
4. Source and destination PCs

23. CDMA stands for \_\_\_\_\_.

1. Carrier Division Multiple Access
2. Carrier Data Multiple Access
3. Code Data Multiple Access
4. Code Division Multiple Access

24. If users must take turns using the link, then it is called Spatial Sharing.

1. true
2. false

25. **GO-BACK-N** protocol is one of the \_\_\_\_\_ layer protocols, which is used to control the flow of data during the transmission.

1. Physical
- 2. Network**
3. Presentation
4. Data link

26. In \_\_\_\_\_ a dedicated communication path is established between two stations through the nodes of the network

- 1. Circuit switching**
2. Packet switching

27. Signals travel through fiber optic cable are in the form of \_\_\_\_\_.

- 1. Light**
2. Bits
3. Electromagnetic
4. Bytes

28. \_\_\_\_\_ contains a repeater.

- 1. Active hub**
2. Passive hub

29. In frequency shift keying, \_\_\_\_\_ remain(s) constant.

1. Frequency
- 2. Amplitude**

3. Both Amplitude and Phase

**4. Both Phase and Frequency**

**30. \_\_\_\_\_ is/are un-reliable protocol(s).**

1. IP and TCP

2. UDP

**3. IP and UDP**

4. TCP

**31. \_\_\_\_\_ is not a function of Data Link Control.**

1. Framing

**2. Modulation**

3. Error Control

4. Flow Control

**32. Each computer on the Internet has a unique numeric address called a(n) \_\_\_\_\_.**

1. domain address

2. protocol address

**3. IP address**

4. Web address

**33. dB is negative if a signal is \_\_\_\_\_.**

**1. Attenuated**

2. Distorted

3. Amplified

4. Noisy

**34. We can have \_\_\_\_\_ different groups with a 4 bit block.**

1. 4

**2. 8**

3. 16

4. 32

**35. Line Configuration refers to the way two or more devices attach to a \_\_\_\_\_.**

1. Path

2. Circuit

**3. Link**

4. Router

**36. Error detection and correction are the services provided by \_\_\_\_\_ layer.**

**1. Data link layer**

2. Application layer

3. Physical layer

4. Session layer

**37. Which of the following sublayer resolves the contention for the shared media.**

**1. MAC**

2. LLC
3. Physical
4. None of the given.

**38. There are \_\_\_\_\_ basic categories of multiplexing.**

1. 2
2. 3
3. 7
4. 5

**39. Low pass channel has \_\_\_\_\_ bandwidth between two stations.**

1. Dedicated
2. Shared
3. Multiplexed
4. Infinite

**40. BSC supports \_\_\_\_\_ transmission using \_\_\_\_\_ flow control.**

1. Full-duplex, Stop and wait ARQ
2. Half-duplex, Selective reject ARQ
3. Full-duplex, Selective reject ARQ
4. Half-duplex, Stop and wait ARQ

**41. Quantization is a method of assigning \_\_\_\_\_ values in a specific range to sampled instances.**

1. Negative

2. Positive

3. Binary

**4. Integral**

**42. \_\_\_\_\_ encoding is almost obsolete today**

1. Bipolar

**2. Unipolar**

3. Polar

4. None of the given

**43. YMODEM has \_\_\_\_\_ Byte of data unit.**

**1. 1024**

2. 256

3. 128

4. 512

**44. Like Data link layer, \_\_\_\_\_ is also responsible for Flow control.**

**1. Transport Layer**

2. Session Layer

**45. In CRC there is no error if the remainder at the receiver is \_\_\_\_\_.**

1. Equal to the remainder at the sender

**2. Zero**

3. Nonzero

4. The quotient at the sender

**46. \_\_\_\_\_ is normally used where speed is priority in data transfer.**

**1. Serial data transmission**

2. Hybrid data transmission

3. Parallel data transmission

4. Both parallel and serial data transmission

**47. In Block coding scheme, number of code words is always \_\_\_\_\_ data words.**

1. Less than

**2. Equal to**

3. Greater than

4. Same as

**48. A portion of the path that carries TX between a given pair of devices is known as \_\_\_\_\_.**

1. Direct Point

2. Bridge

**3. Channel**

4. Access Point

49. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?

1. 100

2. 500

3. 300

4. 600

50. \_\_\_\_\_ is the process of converting digital data to digital signals.

1. CSMA/CD

2. CSMA/CA

3. Line coding

4. Segmentation

51. \_\_\_\_\_ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

1. Fast Ethernet

2. Standard Ethernet

3. Gigabit Ethernet

4. 10BASE-T Ethernet

52. Congestion Control is a feature of layer(s).

1. Data Link

2. Network

3. Transport

**4. All mentioned**

**53. In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.**

- 1. Same as
- 2. Two times**
- 3. Three times
- 4. Four times

**54. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 300, 600, 900 and 1200?**

- 1. 100
- 2. 900**
- 3. 300
- 4. 600

**55. \_\_\_\_\_ signal can take infinite levels of intensity over time.**

- 1. Digital
- 2. Discrete
- 3. Analog**
- 4. Logical

**56. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of \_\_\_\_\_.**

- 1. Local Area Network
- 2. Metropolitan Area Network**

3. Wide Area Network

4. Home Based Network

**57. Which layer is concerned with the syntax and semantics of info exchange between two systems?**

1. Physical

2. Application

3. Session

**4. Presentation**

**58. Time domain shows changes in \_\_\_\_\_.**

1. time w.r.t signal amplitude

**2. signal amplitude w.r.t time**

3. time

4. All of the given

**59. In ASK, Noise usually affects the \_\_\_\_\_.**

**1. Amplitude**

2. Phase

3. Sample

4. Frequency

**60. In analog transmission, the base signal is called the \_\_\_\_\_.**

**1. Carrier Signal**

2. Analog signal

3. Digital signal

4. Modulated signal

**61. Which of the following is most affected by noise?**

1. PSK

**2. ASK**

3. FSK

4. QAM

**62. In Synchronous TDM, Time slots are not same in number as input devices.**

1. True

**2. False**

**63. The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.**

1. Duplicated

2. Blocked

3. Cleaned

**4. Escaped**

**64. Guard bands are used in \_\_\_\_\_ multiplexing technique, to avoid overlapping of frequency bands assigned to each user.**

1. PDM

2. CSMA

3. TDM

#### 4. FDM

65. DLC in Data Link Layer stands for \_\_\_\_\_.

1. Divide Line Communication
2. Data Line Code
3. Data Link Control
4. Demand Link Coordination

66. In Pulse Code Modulation (PCM), the sampling is dependent on \_\_\_\_\_.

1. Time
2. Amplitude
3. Frequency
4. Signal Rate

67. Digital signals are referred to be \_\_\_\_\_.

1. Continuous
2. Discrete
3. All of the given
4. None of the given

68. Two or more computers connected so that they can communicate with each other and share information is called a \_\_\_\_\_.

1. satellite
2. protocol

3. broadcast

4. network

69. Which error detection method uses one's complement arithmetic.

1. Cyclic Redundancy Check (CRC)

2. Longitudinal Red Check (LRC)

3. Checksum

4. Simple Parity Check

70. WDM stands for \_\_\_\_\_.

1. Wave Division Multiplexing

2. Wrong Division Multiplexing

3. Well Division Multiplexing

4. West Division Multiplexing

71. Analog refers to something that is continuous in \_\_\_\_\_.

1. Time

2. Space

3. Frequency

4. None of the given

72. What is the frequency of a sine wave if it completes one cycle in 5 seconds?

1. 0.5

2. 0.6

3. 0.3

4. 0.2

**73. In an analog hierarchy to carry voice channels, a group can carry \_\_\_\_\_ voice channels.**

1. 60

2. 12

3. 20

4. 10

**74. STP is more expensive than UTP and is less susceptible to noise.**

1. True

2. False

**75. Switch is a network device which operates on the \_\_\_\_\_ layer of the TCP/IP protocol suite.**

1. Application

2. Transport

3. Presentation

4. Data link

**76. \_\_\_\_\_ are used to exchange session management and control information between connected devices.**

1. I-frames

2. U-frames

3. S-frames

4. N-frames

77. The message 1110 sent by a source is received by a destination as 1011. This is \_\_\_\_\_ type of error.

1. Burst

2. Single-Bit

3. Uni-Bit

4. Hamming



# **Cs601 Mid term**

**Topic 1 to 113 cover in this quiz**

**Orange Monkey Team**

1. At the application layer, object/information is in the form of \_\_\_\_\_.  
Message...confirm

2. \_\_\_\_\_ is the first step in PCM technique  
Sampling...confirm

3. Communication between two user in a cellular (Mobile) network is an example of \_\_\_\_ communication  
Full duplex....confirm

4. Transmission medium are of \_\_\_\_\_ types.  
Two....confirm

5. A frequency is called \_\_\_\_\_ if sine wave changes with time  
Infinte frequency...confirm

6. A frequency is called \_\_\_\_\_ if sine wave does not change with time  
Zero frequency ....confirm

7. Open system interconnection (OSI) has \_\_\_\_ number of layers.  
7...confirm

8. In unidirectional antenna, the common point where all the lines intersect each other is \_\_\_\_.

Focus...confirm

9. To calculate the data rate for noisy channel \_\_\_\_ formula is used.

Shannon....confirm

10. Television broadcast where user can only view the information sent in the form of video is an example of \_\_\_\_ communication.

Simplex

11.  $N = S \cdot r$  is the formula for finding

Baud rate.....confirm

12. TCP/IP protocol suit has \_\_\_\_ layers.

5....confirm

13. Unit of phase is \_\_\_\_.

Degree....confirm

14. For a noiseless channel theoretical formula to calculate the data rate was developed by.

Nyquist.....confirm

15. In context of bandwidth-delay product the cross section of the pipe represents the \_\_\_\_.

Bandwidth...confirm

16. Uni polar Polar and Bipolar are types of \_\_\_\_ coding scheme.

Line...confirm

17. In \_\_\_\_ scheme the voltages are on both sides of the time axis and voltage level for 0 can be positive and the voltage level for 1 can be negative.

**Polar....confirm**

**18.In \_\_\_\_\_ transmission medium start and stop bits are used**

**Asynchronous....confirm from net**

**19.In layered architecture of network logical connection between layers means.**

**They are connected in peers.....confirm**

**20.OSI stands for \_\_\_\_\_.**

**Open system interconnection....confirm**

**21.Quadrature Amplitude Modulation is the mechanism of \_\_\_\_\_ conversion.**

**Digital to analog...confirm**

**22.Writing used for transmission modes depends upon \_\_\_\_\_.**

**Data stream...confirm**

**23.The maximum bandwidth required for QAM transmission is the same as that required for \_ and \_\_\_\_\_ transmission.**

**ASK, PSK**

**24.In optical fiber cable the ratio of the diameter of their core to the diameter of their cladding, both expressed in \_\_\_\_\_.**

**Micrometers...confirm**

**25.Radio waves are propagated in \_\_\_\_\_ whenever they are transmitted by an antenna.**

**All direction ...confirm from net**

**26.\_\_\_\_\_ component of data communication governs the overall communication on the network.**

**Transmission medium...confirm**

27. In binary ASK, the peak amplitude of one signal level is 0 and the other is the same as the \_\_\_\_\_.

Amplitude

28. In \_\_\_\_\_ scheme, all the signal levels are on one side of the time axis, either above or below.

Unipolar...confirm from net

29. Simple sine wave can be used to send.

Amplitude

30. Paging system in data communication uses \_\_\_\_\_ waves.

Radio waves...confirm from net

31. Multipoint channel can be shared in \_\_\_\_\_ manner.

Both temporal and spartial...confirm

32. \_\_\_\_\_ is an example of multipoint topology

Bus...confirm

33. Bandwidth can be measured in \_\_\_\_\_.

Bps...confirm

34. In \_\_\_\_\_ encoding scheme, we use three levels positive negative and zero

Bipolar...confirm from net

35. Header size \_\_\_\_\_ while moving down the layers at sending side

Decreases

36. \_\_\_\_\_ signals have limited number of discrete value

Digital....confirm

37. \_\_\_\_\_ is a block coding method

**NRZ-1....confirm**

**38. Technique that allows the simultaneous transmission of multiple signals across a single data link, is known as \_\_\_\_\_.**

**Multiplexing....confirm**

**39. Simultaneous transmission of multiple signals across a single data link is called.**

**Multiplexing...confirm**

**40. Minimum bandwidth required for Manchester line coding scheme is \_\_\_\_\_.**

**1Mhz...confirm**

**41. AMI is a popular \_\_\_\_\_ encoding method.**

**Bi-polar....confirm**

**42. TCP/IP was presented by \_\_\_\_\_ in a research paper**

**Vin Cerf and bob Kahn...confirm**

**43. Among the following cables given below \_\_\_\_\_ cable has the cost as compared to others.**

**Medium....confirm**

**44. Among the following cables given below \_\_\_\_\_ cable has the lowest cast as compared to others.**

**Coaxial cable...confirm**

**45. Among the following cables given below \_\_\_\_\_ cable has the highest cast as compared to others.**

**Fiber optic cable...confirm**

46. \_\_\_\_\_ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

**n-1....confirm**

47.A twisted pair cable consists of \_\_\_\_\_ conductors

**2 ....confirm**

48.Packets travel in \_\_\_\_\_ layer of TCP/IP protocol suit

**Network ....confirm from net**

49.HDB3 falls under which coding scheme?

**Bipolar...confirm**

50.According to the Nyquist theorem, the sampling rate must be at least \_\_\_\_\_ times the highest frequency contained in the signal.

**2.....confirm**

51. \_\_\_\_\_ protocol suit is being used by the modern internet communication

**TCP/IP...confirm**

52.LF(low frequency) band is used in \_\_\_\_\_ propagation

**Ground....confirm**

53. \_\_\_\_\_ is a type of serial transmission in which data bits are transmitted as a continuous stream in time with a master clock, start bits, stop bits and gaps are not used. Transmitter and receiver time is sync.

**Synchronous....confirm**

54.Identify the correct formula to calculate Propagation time.

**Propagation time= Distance/ propagation speed....confirm**

55.The communication of the application, transport and network layer is \_\_\_\_\_

Physical...confirm

56.Frames travel in \_\_\_\_\_ layer of TCP/IP Protocol suite.

Data link

57.Manchester encoding scheme achieves the same level of synchronization as \_\_\_\_\_.

RZ.....confirm

58.In Manchester and differential Manchester encoding schemes, the signal rate is \_\_\_\_\_ than that of NRZ.

Double...confirm from net

59.In a bidirectional communication each layer performs two \_\_\_\_\_ tasks in each direction.

Opposite...confirm

60.\_\_\_\_\_ is composed of thousands of interconnected networks expanded over large geographical area.

Local area network

61.Time taken by a periodic signal to complete one cycle is called.

Period...confirm from net

62.There are \_\_ common scrambling technique

Two ....confirm

63.The bipolar encoding scheme was developed as an alternative of \_\_\_\_\_.

NRZ...confirm from net

64.FSK stands for \_\_\_\_\_.

**Frequency shift keying....confirm**

**65. In \_\_\_ encoding scheme, the voltage level oscillates between a positive and a negative value although it may remain at zero level between the two values.**

**Polar...confirm from net**

**66. \_\_\_\_\_ is an example of transmitter.**

**Modem**

**67. The material used for conduction in twisted pair cable is \_\_\_\_\_.**

**Copper**

**68. RG-11 having impedance of 50 ohm is used in \_\_\_\_\_ Ethernet**

**Thin....confirm**

**69. \_\_\_\_\_ encoding scheme is somewhat a combination of the RZ and NRZ-I scheme**

**Manchester encoding...confirm from net**

**70. \_\_\_\_\_ has less electromagnetic interference as compared to other guided medium.**

**Unshielded Twisted Pair**

**71. Electromagnetic waves ranging in frequencies between \_\_\_\_\_ are called microwaves.**

**1 to 300 GHz**

**72. Switch is a \_\_\_\_\_ layer devices**

**Data link...confirm**

73. \_\_\_\_\_ is Shannon capacity formula to determine highest theoretically data rate for a noisy channel.

Capacity = Bandwidth \*  $\log_2(1+SNR)$ ....confirm

74. \_\_\_\_\_ is the range of frequencies contained in a composite signal.

Bandwidth...confirm

75. In PCM technique Quantization is an \_\_\_\_\_ process

Un-balanced

76. In Ring topology, each node has a point to point dedicated link with.

Exactly 2 nodes....confirm

77. In \_\_\_\_\_ modulation process is used by the transmitting devices to shift their data to a specific carrier frequency for multiplexing.

Fdm...confirm

78. Five stations will have \_\_\_\_\_ full duplexed dedicated links among them to be connected in a Mesh topology.

10....confirm

79. Which of the following technique(s) is used for Analog-to-Digital Conversion.

Both Pulse Code Modulation(PCM) and Delta Modulation(DM)...confirm

80. The human voice normally frequency range from \_\_\_\_\_.

0 to 3000Hz

81. The layer of TCP/IP protocol suite which is on the top is known as \_\_\_\_\_ layer.

Application...confirm

82. In \_\_\_\_\_ type of error, two or more bits in the data unit will be changed.

Burst....confirm

83. \_\_\_\_\_ is the process of converting digital data to digital signal.

Line coding.....confrim

84. Manchester encoding scheme uses and inversion at the \_\_\_\_\_ of each bit.

Middle....confirm

85. FF:FF:FF:FF:FF:FF is a \_\_\_\_\_ link layer address.

Broadcast.....confrim

86. A3 34 45 11 92 F1 is a \_\_\_\_\_ link layer address.

Data, unicast....confirm

87. High-Level Data Link Control(HDLC) defines \_\_\_\_\_ types of frames to deal with the transfer modes.

3.....confirm

88. \_\_\_\_\_ is the number of bits sent in 1 second

Bit rate(bps)...confirm

89. \_\_\_\_\_ Time required to send 1 single bit.

Bit interval...confirm

90. \_\_\_\_\_ are used to transport user data and control information relating to user data

Information frams(I-frames).....confirm

91. \_\_\_\_\_ are used only to transport control information

Supervisory frames(S-frames).....confirm

92. \_\_\_\_\_ are reserved for system Management

Unnumbered frames(U-frames).....confirm

93. \_\_\_\_\_ are used to exchange session management and control information between connected devices.

U-frames....confrim

94. In order to find Hamming distance between two pair of words \_\_\_\_\_ logical operation is used.

XOR....confrim

95. Data Link layer receives services from the \_\_\_\_\_ layer.  
Physical

96. The \_\_\_\_\_ is the number of signal elements sent per unit time.

Bit rate.....confrim

97. In order to resolve link layer address of the next node, the first generated ARP request is always in \_\_\_\_\_ mode.

Broadcast

98. In Block coding we divide our message into blocks of k bits called as \_\_\_\_\_.

Data words.....confirm

99. To control and manage the transfer of data, a protocol called point to point (PPP) is used at the \_\_\_\_\_ layer

Data Link

100. In character oriented protocol, each character of frame is encoded using \_\_\_\_\_ coding system.

ASCII

101. Error detection and correction are the services provided by \_\_\_\_\_ layer.

Data link layer ....confirm from net

102. \_\_\_\_\_ is used in the stop-and-wait protocol to overcome the issue of duplication.

Frame sequencing and Acknowledgment numbering...confirm from net

103. Stop-and-wait protocol is used for \_\_\_\_\_.

Both flow and error control....confirm

104. Error correction is more \_\_\_\_\_ than the error detection  
Difficult...confirm from net

105. Payload field of the PPP frame is of variable size and it can carry maximum data up to \_\_\_\_\_.

1500 bytes....confirm from net

106. If duration of noise is decreased during a transmission over noisy channel. Fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise.

More....confirm from net

107. One of the most common protocols for point-to-point access is \_\_\_\_\_.

PPP...confirm

108. In linear Block codes \_\_\_\_\_ of two valid code words creates another valid code word.

XOR...confirm

109. Which layer is concerned with the syntax and semantics of info exchange between two systems?

### **Presentation**

**110. Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching**

**false**

**111. In OSI model headers are added at layer \_\_\_\_\_.**

**6,5,4,3,2 only**

**112. \_\_\_\_\_ works with topologies in which one device is designated as a primary station and the other devices are secondary stations.**

**Polling**

**113. Signals travel through fiber optic cable are in the form of \_\_\_\_\_.**

**Light...confirm from net**

**114. PSK is susceptible to the noise degradation.**

**False**

**115. What is the frequency of a sine wave if it completes one cycle in 5 seconds?**

**0.2**

**116. Which one is not the function of data link layer?**

**Line discipline**

**117. Error detection and correction are the services provided by \_\_\_\_\_ layer.**

**Data link layer**

118. \_\_\_\_\_ is not a function of Data Link Control.

**Modulation**

119. No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between \_\_\_\_\_.

**Two adjacent nodes**

120. The inversion of the level represents a 1 bit in \_\_\_\_\_ encoding

**NRZ-I**

121. Guard bands are used in \_\_\_\_\_ multiplexing technique, guard bands to avoid overlapping of frequency bands assigned to each user.

**FDM...confirm**

122. The \_\_\_\_\_ protocol uses both flow and error control.

**Stop-and-Wait**

123. Each computer on the Internet has a unique numeric address called a(n) \_\_\_\_\_.

**IP address**

124. WDM stands for \_\_\_\_\_.

**Wave Division Multiplexing**

125. A fiber -optic cable transmit signals in the form of \_\_\_\_\_.

**Light**

126. CDMA stands for \_\_\_\_\_.

**Code Division Multiple Access**

127. Quantization is a method of assigning \_\_\_\_\_ values in a specific range to sampled instances.

Integral

128. STP is more expensive than UTP and is less susceptible to noise.

True

129. In circuit switched networks we have low efficiency but minimal \_\_\_\_\_.

Delay....confirm

130. Which level of the TCP/IP reference model routes data/information across a network channel?

Network Layer

131. A portion of the path that carries TX between a given pair of devices is known as \_\_\_\_\_.

Channel

132. Digital signals are referred to be \_\_\_\_\_.

Discrete

133. BSC supports \_\_\_\_\_ transmission using \_\_\_\_\_ flow control.

Half-duplex, Stop and wait ARQ

134. \_\_\_\_\_ is the rate of change with respect to time.

Amplitude

135. The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?

Standard Ethernet

136. If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error locations.

16

137. In Pulse Code Modulation (PCM), the sampling is dependent on \_\_\_\_\_.

Amplitude...confirm

138. The extra bits added with the original data for error detection/correction are called \_\_\_\_\_.

Redundant bits

139. Which of the following sublayer resolves the contention for the shared media.

MAC

140. PAP and CHAP are used by \_\_\_\_\_ as Authentication Protocols

PPP...confirm

141. Media Access Control (MAC) and Data link (DLC) constitute \_\_\_\_\_ layer

Data Link....confirm

142. In \_\_\_\_\_, each station sends a frame whenever it has a frame to send.

Slotted ALOHA

143. What is the period of a Sine wave having frequency of 10 Hz?

0.1

144. Is there any difference between Data communication and telecommunication are network?

No

145. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as \_\_\_\_\_.

Packetizing

146. \_\_\_\_\_ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

$N-1$

147. Following characteristics fully describe \_\_\_\_\_,  
Amplitude  
Period/Frequency  
Phase

Sine Waves

148. What is the period of a Sine wave having frequency of 5 Hz?

0.2

149. The inversion of the level at 1 bit is called as \_\_\_\_\_.

NRZ-I

150. In \_\_\_\_\_, both peak amplitude and frequency remains constant as the phase changes.

PSK

151. At the CRC generator, \_\_\_\_\_ added to the data unit before the divisional process.

0s are

152. In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.

Two times

153. In Block coding we divide our message into blocks of k bits called as \_\_\_\_\_.

Data words

154. Analog refers to something that is continuous in \_\_\_\_\_.

Time

155. \_\_\_\_\_ of a signal is the collection of all the component frequencies it contains.

Frequency spectrum

156. \_\_\_\_\_ encoding is almost obsolete today

Unipolar

157. Which of the following is most affected by noise?

ASK

158. In \_\_\_\_\_, a signal is directed straight from antenna to antenna.

Line of sight propagation

159. To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the \_\_\_\_\_ layer.

## Network

160. The level of signal is always positive in NRZ encoding.

false

161. \_\_\_\_\_ signal can take infinite levels of intensity over time.

Analog....confirm from net

162. \_\_\_\_\_ is sometimes called the bit rate.

Data rate...confirm from net

163. \_\_\_\_\_ contains a repeater.

Active hub

164. \_\_\_\_\_ is/are un-reliable protocol(s).

UDP

165. Low pass channel has \_\_\_\_\_ bandwidth between two stations.

Shared...confirm from net

166. The message 1110 sent by a source is received by a destination as 1011. This is \_\_\_\_\_ type of error.

Burst

167. TROPOSHERE is the layer of atmosphere above the troposphere but below space

False

168. In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.

Two times

169. Asynchronous transmission is \_\_\_\_\_.

Slow

170. Radio is an example of \_\_\_\_\_ signal conversion.

Analog to analog

171. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?

Longitudinal Red Check (LRC)

172. Session layer is responsible for \_\_\_\_\_.

Maintaining the connection

173. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise.

More

174. S-frames in High-Level Data Link Control (HDLC) are only used to transfer \_\_\_\_\_ information.

Control

175. In ASK, both \_\_\_\_\_ and \_\_\_\_\_ remain constant.

Frequency, Phase

176. Encoding digital data into digital signals is called \_\_\_\_\_.

Digital-to-Digital Conversion

177. GO-BACK-N protocol is one of the \_\_\_\_\_ layer protocols, which is used to control the flow of data during the transmission.

Network

178. In ASK correct formula for calculating the bandwidth is as \_\_\_\_\_.

$B=(1+d)S$ ....confirm

179. The type of ALOHA which improves the efficiency of Pure ALOHA is \_\_\_\_\_.

Slotted ALOHA

180. In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

Polling

181. The \_\_\_\_\_ generates the data and passes it along with any control information to a \_\_\_\_\_.

DTE, DCE

182. \_\_\_\_\_ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

Frame sequencing and Acknowledgement numbering

183. Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol.

Network

184. The Internet is \_\_\_\_\_.

a global network of computers networks

185. We can have \_\_\_\_\_ different groups with a 4 bit block.

8

186. Analog To Analog Conversion Methods include \_\_\_\_\_.

**AM, PM and FM**

**187. DLC in Data Link Layer stands for \_\_\_\_\_.**

**Data Link Control**

**188. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?**

**600**

**189. \_\_\_\_\_ category of coaxial cable is used for thin Ethernet.**

**RG-58....confirm**

**190. \_\_\_\_\_ category of coaxial cable is used for thin Ethernet.**

**RG-8, RG-9, RG-11 .....confirm**

**191. \_\_\_\_\_ category of coaxial cable is used for thin Ethernet.**

**RG-8, RG-9, RG-11 .....confirm**

**192. \_\_\_\_\_ category of coaxial cable is used for Cable TV**

**RG59....confirm**

**193. In \_\_\_\_\_, each station is allocated a time slot during which it can send data.**

**TDMA**

**194. \_\_\_\_\_ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.**

**10BASE-T Ethernet**

**195. Time domain plot show changes in signal phase with respect to time.**

false

196. dB is negative if a signal is \_\_\_\_\_.

Attenuated

197. Which of the following is not a characteristic of a sine wave?

Segmentation...confirm

198. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 300, 600, 900 and 1200?

900

199. Asynchronous TDM is efficient only when the size of the time slot is kept relatively \_\_\_\_\_.

Large

200. A sine wave is defined by \_\_\_\_\_ characteristics.

3

Switch is a network device which operates on the \_\_\_\_\_ layer of the TCP/IP protocol suite.

Data link

201. In even parity check code, the value of syndrome is \_\_\_\_\_ if the number of 1s is even.

0

202. 4Collisions in CSMA/CA are avoided through \_\_\_\_\_.

All of the given

203. Human voice is example of \_\_\_\_\_ signal.

## Analog

204. Time domain shows changes in \_\_\_\_\_.

signal amplitude w.r.t time

205. Normally the value of Kmax in pure ALOHA is \_\_\_\_\_.

15

206. Central Hub in a Tree is an \_\_\_\_\_.

Active hub

207. In selective-reject ARQ, only the specific damaged or lost frame is \_\_\_\_\_.

Retransmitted

208. If users must take turns using the link, then it is called Spatial Sharing.

false

209. At the CRC checker, \_\_\_\_\_ means that the data unit is damaged.

A nonzero remainder

210. Digitizing an Analog Signal is called \_\_\_\_\_.

Analog-to-Digital Conversion

211. WDM stands for \_\_\_\_\_.

Wavelength Division Multiplexing

212. What was the main draw back in ALOHA Random-Access method?

All of the given

213. YMODEM has \_\_\_\_\_ Byte of data unit.

1024

214. Like Data link layer, \_\_\_\_\_ is also responsible for Flow control.

a. Transport Layer

215. In Block coding scheme, number of code words is always \_\_\_\_\_ data words.

Equal to

216. RZ stands for \_\_\_\_\_.

Return of Zero

217. Line Configuration refers to the way two or more devices attach to a \_\_\_\_\_.

Link

218. The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

Escaped

219. At the Application layer, object/information is in the form of \_\_\_\_\_.

Packet

220. The term \_\_\_\_\_ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

## Single Bit Error

221. What is the frequency of a sine wave if it completes one cycle in 2 seconds?

0.5

222. POST stand for \_\_\_\_\_?

Plain Old Telephone Service

223. To make sure that source does not overwhelm destination by sending data faster than it can be handled and processed is called?

Flow Control

224. If a digital signal has "L" number of levels. \_\_\_\_\_ is the number of bits required to represent each level.

Log base 2 L....confirm

225. In ASK, Noise usually affects the \_\_\_\_\_.

Amplitude

226. In order to find the location of two errors in an eight bit stream, we have to see \_\_\_\_\_ different combinations.

28

227. Which one of the following is not a Channelization Protocol?

CSMA

228. Which one of the following is a correct-access method?

### Reservation

229. By using \_\_\_\_\_ system, change in the file contents during file transmission can be avoided.

### Error Detection and Correction

230. Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.

Five

231. A broadcast signal is received by the orbiting satellites which rebroadcasts \_\_\_\_\_.

none of the given

232. What is the period of a Sine wave having frequency of 2 Hz?

0.5

233. \_\_\_\_\_ mode of serial transmission guarantees fixed rate of data.

Isochronous...confirm from net

234. Time division multiplexing is used in \_\_\_\_\_ systems.

Digital....confirm from net

235. The most common type of connector used by coaxial cable is \_\_\_\_\_.

BNC

236. Which of the following statement is correct?

Digital signals are less prone to Noise

237. In an analog hierarchy to carry voice channels, a group can carry \_\_\_\_\_ voice channels.

12(4kHz)....confirm

238. In an analog hierarchy to carry 12 voice channels, 5 group \_\_\_\_\_

48kHz....confirm

239. In an analog hierarchy to carry 60 voice channels, a 10 super group \_\_\_\_\_ bandwidth

240kHz....confirm

240. In an analog hierarchy to carry 600 voice channels, a 6 Master group \_\_\_\_\_

2.52Mhz....confirm

241. In an analog hierarchy to carry 3600 voice channels, a Jumbo group \_\_\_\_\_

16,984MHz .....confirm

242. In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_ frame.

EOT

243. Data is transmitted using light through a \_\_\_\_\_ cable.

fiber-optic

244. Data Link layer provides services to the \_\_\_\_\_ layer.

Network

245. If the Hamming distance between sent and received code word is \_\_\_\_\_, then it shows that received data is corrupted.

$\neq 0$

246. \_\_\_\_\_ is responsible for governing node to node communication.

Data Link layer

247. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of \_\_\_\_\_.

Metropolitan Area Network

248. In frequency shift keying, \_\_\_\_\_ remain(s) constant.

Both Peak amplitude and Phase ...confirm

249. In Amplitude shift keying, \_\_\_\_\_ remain(s) constant.

Both frequency and phase...confirm

250. In Phase shift keying, \_\_\_\_\_ remain(s) constant.

Both peak amplitude and frequency...confirm

251. Analog signals are refers to be \_\_\_\_\_

Continuous

252. Baud rate is greater than or equal to the bit rate.

False

253. In statistical time division multiplexing, number of slots in frame are less than \_\_\_\_\_.

Input lines.....confirm

254. \_\_\_\_\_ are used to exchange session management and control information between connected devices.

U-frames

255. Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as \_\_\_\_\_

DTE

256. There are three types of Hubs that exist. Which of the following options correctly describes these three?

Passive, Active, Intelligent

257. Which of the following consist of just one redundant bit per data unit?

Simple Parity Check

258. In the encoding where positive to negative transition is one and vice versa is zero.

None of given option

259. Error correction is more \_\_\_\_\_ than the error detection.

Difficult

260. If the window size is 63 in go-back N ARQ then what is the range of sequence number?

0 to 63

261. \_\_\_\_\_ is normally used where speed is priority in data transfer.

Hybrid transmission

262. In Synchronous TDM, Time slots are not same in number as input devices.

False

263. In analog transmission, the base signal is called the\_\_\_\_\_.

Carrier Signal

264. In \_\_\_\_\_ a dedicated communication path is established between two stations through the nodes of the network

Circuit switching

265. Which one of the following is a Random-Access protocol?

CDMA/CD

266. A \_\_\_\_\_ modem is used to connect two DTEs directly.

NULL

267. The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

Escaped

268. \_\_\_\_\_ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

PAP

269. In CRC there is no error if the remainder at the receiver is \_\_\_\_\_.

Zero

270. VCI address is \_\_\_\_\_ address in virtual circuit approach.

Global...confirm from net

271. \_\_\_\_\_ is an Authentication Protocol, which uses two-step process to authenticate user information.

PAP

272. Possession of TOKEN gives the station the right to \_\_\_\_\_.

Send the data

273. The transmission medium that carries the message is referred to as the \_\_\_\_\_.

communication channel

274. Latency is made of \_\_\_\_\_ components.

Four....confirm

275. Sampling means measuring \_\_\_\_\_ of signal at equal intervals

Amplitudes

276. Which error detection method uses one's complement arithmetic.

Checksum

277. To calculate the data rate for noisy channel \_\_\_\_\_ formula is used.

Shannon.....confirm

278. To calculate the data rate for noiseless channel \_\_\_\_\_ formula is used.

Nyquist....confirm

279. Checksum is an error-detection technique that can be applied to a message of \_\_\_\_\_ length.

Any

280. In \_\_\_\_\_ each band is reserved for a specific station, and it belongs to the station all the time.

FDMA

281. Flow control is needed to prevent \_\_\_\_\_.

Overflow of the receiver buffer

282. Quadrature Amplitude Modulation (QAM) is the combination of \_\_\_\_\_ and \_\_\_\_\_.

FSK, PSK

283. Like 10 Base 5, 10 Base 2 is a \_\_\_\_\_ topology LAN.

Bus

284. In the \_\_\_\_\_ method, the stations in a network are organized in a logical ring.

Token-passing

285. Two or more computers connected so that they can communicate with each other and share information is called a \_\_\_\_\_.

network

286. The logical connection between the peer layers is \_\_\_\_\_ connection.

Direct....confirm from net

287. In CSMA/CA, \_\_\_\_\_ is the amount of time divided into slots.

Contention Window

288. \_\_\_\_\_ relative measures the strength of two signals.

Decibal ....confirm

289. A sine wave is \_\_\_\_\_.

periodic and continuous

290. There are \_\_\_\_\_ types of guided media.

3

291. AM signal requires \_\_\_\_\_ the BW of original signal

Twice

292. CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to authenticate user information.

Three-way

293. There are \_\_\_\_\_ types of serial transmission.

2

294. There are \_\_\_\_\_ basic categories of multiplexing.

3

295. Congestion Control is a feature of layer(s).

All mentioned

296. Most commonly used connector for twisted pair cable is \_\_\_\_\_.

RJ-45....confirm from net

297. \_\_\_\_\_ does not happen until all the data is transferred and all of the resources are freed.

Switching

298. A local telephone network use \_\_\_\_\_ network.

Circuit switched...confirm from net

299. Microwaves are \_\_\_\_\_.

Unidirectional ....confrim

300. \_\_\_\_\_ have established standards for using these signals for communication between devices such as keyboards, mice, PCs and printers

Infrared waves...confirm from net

301. Datagram approach and virtual circuit approach are two popular approaches which lead us to \_\_\_\_\_.

Circuit switching....confirm

302. In circuit switching when two end devices 'A' and 'B' connects with each other they send \_\_\_\_\_ request.

Acceptance

303. In optical fiber a glass or plastic core is surrounded by a \_\_\_\_\_.

Cladding...confirm

304. \_\_\_\_\_ are used for multicast communication, such as radio and television.

Radio waves...confirm

305. DLC in data link layer stands for \_\_\_\_\_

Data Link control....confirm

306. In virtual Circuit network all packets follow \_\_\_\_\_ path

Dedicated

307. Multi mode technology has \_\_\_\_\_ main types for light propagating.

Two

308. If the message is going to pass through a packet-switched network. It can be divided into packets of \_\_\_\_\_.

Both fix and variable size

309. According to state, fewer than \_\_\_\_\_ crossbar points can be used at once.

25%

310. \_\_\_\_\_ and \_\_\_\_\_ are the two types of addressing in virtual circuit approach.

Local, Global

311. Cable TV networks use \_\_\_\_\_ cables

Coaxial....confirm from net

312. The size of the packet is determined by the network and \_\_\_\_\_.

Governing Protocol

313. In circuit switching total delay is combination of connection time. Data transfer time and \_\_\_\_\_

Connection tear down time

314. Which one of the following is not among the required phases for the actual communication in a circuit-switched network?

Protocol

315. Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called \_\_\_\_\_.

Microwaves

316. If we increase the data rate of a transmission over noisy channel, fixed noise duration will cause \_\_\_\_\_ number of bits being impacted by the noise.

Less

317. In case of higher data rate, the number of impacted bits will be \_\_\_\_\_.

Higher

318. In baseband transmission, a digital signal is transmitted as \_\_\_\_\_.

Analog signal....confirm from net

319. \_\_\_\_\_ Cable is used in wavelength division multiplexing.

Fiber optic....confirm from net

320. The \_\_\_\_\_ technique expands the bandwidth of a signal by replacing each data bit with n bits using a spreading code.

DSSS...confirm from net

321. When the bandwidth of a link is greater than combined bandwidths of the

Frequency division....confirm from net

322. According to the Fourier analysis, frequencies obtained after decomposition of no period.

Continuous.....confirm from net

323. Router is a network device which operates on the \_\_ layer of the TCP/IP protocol.

Network...confirm from net

324. Uni polar, polar and bipolar are the types of \_\_\_\_\_

NRZ-I....confirm from net

325. According to stats, lower than \_\_\_\_\_ crossbar point can be used at once .

25%....confirm from net

326. In \_\_\_\_\_ TDM, slots are dynamically allocated to the connected stations to improve bandwidth efficiency.

Statistical...confirm from net

327. In star based network comprising of four computers and on switch, total number of cables needed will be \_\_\_\_\_

4...confirm from net

328. The conversion of analog sample of the signal into digital form is called \_\_\_\_\_ process.

Sampling and quantizing...confirm from net

329. On a communication link, different channels are separated by unused strips of bandwidth called as \_\_\_\_\_.

Guard band

330. \_\_\_\_\_ is an example of host in a network.

Computer...confirm from net

331. In \_\_\_\_\_ TDM, each input connection has an allotment in the output even if it is not sending data.

Synchronous ...confirm

332. \_\_\_\_\_ coding scheme are useful for LAN but not for long ranges

Biphase...confirm

333. In \_\_\_\_\_ signal changes its shape or form.

Distortion...confirm

334. \_\_\_\_\_ Protocol suite is being used by the modern internet communication.

TCP/IP...confirm from net

335. in NRZ-1 the signal is inverted if \_\_\_\_\_ is encountered.

1...confirm from net

336. Simultaneous transmission of multiple signal across a single data link is called \_\_\_\_\_

Multiplexing..confirm from net

337. Two stations are connected via a secure link in which messages are encrypted from source to the destination message is called \_\_\_\_\_

Protocol

Which one of following is a scrambling coding technique?

- B8ZS

According to the Fourier analysis, frequencies obtained after decomposition of digital signals are\_\_\_\_\_.

Discrete

\_\_\_\_\_ is the heart of blocking code.

- Coding

Wireless transmission can be divided into\_\_\_\_\_ broad groups.

Three...confirm from net

Wiring used for transmission modes depends upon\_\_\_\_\_.

- Latency

If a digital transmission system is sending five bits in every half a second, the bit-rate of the system.

- 10 bps...confirm

If the message going to pass through a packet-switched network, it can be divided into packets of\_\_\_\_\_.

- Both fix and Variable size

In an analogue hierarchy to carry voice channels, a group can carry\_\_\_\_\_ voice channels.

12

In optical fiber a glass or plastic core is surrounded by a\_\_\_\_\_.

- Cladding

In transmission impairments \_\_\_\_\_ noise occurs from other source like

Induced...confirm

\_\_\_\_\_ is a digital process that allows connections to share the high bandwidth of a link.

- TDM

Bandwidth in hertz in the range of\_\_\_\_\_ contained in a composite signal.

- Frequencies

LAN stands for\_\_\_\_\_.

• Local Area Network  
Demultiplexer is a \_\_\_\_\_ device.

- One to many

Mr. Asif while sitting in Lahore is linking with his friend in Dubai through Skype is a \_\_\_\_\_.

• Wide Area Network  
In \_\_\_\_\_ modulation, frequency and amplitude remain constant.

• Amplitude  
In TDM based digital hierarchy used by the Telephone companies, DS-1 is a single digital channel of \_\_\_\_\_.

- 1.544 Mbps...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-2 is a single digital channel of \_\_\_\_\_.

6.312 Mbps...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-3 is a single digital channel of \_\_\_\_\_.

44.376MBps...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-4 is a single digital channel of \_\_\_\_\_.

274.176 MBps.....confirm

The bandwidth-delay product defines the number of bits that can fill the \_\_\_\_\_.

- Link

If a digital-signals has four levels, then we need \_\_\_\_\_ bits to represent each level.

- 2

Routing is a function of \_\_\_\_\_ layer.

- Network

\_\_\_\_\_ relatively measures the strength of two signals.

- Decibel

Taking off the header from the message is called \_\_\_\_\_.

- Encryption

In frequency domain plot, which value is replaced with frequency?

- Wavelength

\_\_\_\_\_ is the process of converting binary data to a digital signal.

- Line coding

Low pass channel with \_\_\_\_\_ bandwidth is not real and is used for theoretical modeling.

- Narrow or wide...both are confirm

Unit of phase is \_\_\_\_\_.

- Degree

BFSK stands for \_\_\_\_\_.

- Binary Frequency Sift Key

\_\_\_\_\_ is NOT an example of connecting device.

- TCP

Coaxial cables are categorized by their \_\_\_\_\_ ratings.

Radio Government (RG)....confirm

Which one of the following is not among the required phases for the actual communication in a circuit switched network

- Protocol

A common bipolar encoding scheme is called \_\_\_\_\_.

- AMI

A telephone line analog signal has got the bandwidth of \_\_\_\_\_.

4 kHz...confirm

In \_\_\_\_\_ systems, resources are allocated on demand.

- Circuit switching

A fiber-optic cable transmit signals in the form of \_\_\_\_\_.

- Light

Composite wave can be used to send \_\_\_\_\_.

- Frequency

Routing is the function of \_\_\_\_\_ layer.

- Network

A virtual-circuit network operates on \_\_\_\_\_ layer.

- Datalink

In TCP/IP model, the Physical Layer exchanges data in the form of \_\_\_\_\_.

- Bits...confirm

\_\_\_\_\_ signal completes certain pattern in a specific amount of time.

- Periodic

If a digital signal has four levels, the we need \_\_\_\_\_ bits to represent.

- 2

We can have combination of \_\_\_\_\_ possible codes with a 5-bit code.

- 32

There are \_\_\_\_\_ categories of multiplexing.

- 3

Top most organization that provides support for the Internet Standard process is called \_\_\_\_\_.

- IETF

If the message is going to pass through a packet-switched network, it can be divided into packets of \_\_\_\_\_.

- Both-fix and variable size

\_\_\_\_\_ data moves faster and timing errors are less frequent because the transmitter and receiver time is synced.

Synchronous...confirm

In frequency modulation, the frequency of the oscillator changes according to the \_\_\_\_\_.

- Input Voltage

Binary Amplitude Shift Keying also called as:

- On-Off Keying

Bi-polar uses \_\_\_\_\_ voltage levels.

Two

In serial data transmission \_\_\_\_\_ data transmission mode can be cheap but slower.

- Asynchronous

In star-based network comprising of four computers and one switch, total number of cable needed will be \_\_\_\_\_.

- 4

Parabolic antenna is used for \_\_\_\_\_ communication.

- Microwave...confirm

In circuit switching total Delay is combination of Connection Time and \_\_\_\_\_.

- Connection tear down time

Datagram switching is done on \_\_\_\_\_.

- Network Layer

Token Ring was devised by \_\_\_\_\_.

- IBM...confirm

In TDM based digital hierarchy used by the Telephone companies, DS-0 is a digital channel of \_\_\_\_\_.

- 64 kbps...confirm

\_\_\_\_\_ can be published using Request for Comments (RFCs).

- Internet Draft....confirm

In transmission impairments \_\_\_\_\_ noise occurs from the random motion of electrons in a wire.

- Thermal noise...confirm

Which one of the following is among the required phases for the actual communication in a circuit switched network?

- Data transfer

According to the Fourier analysis, frequencies obtained after decomposition of non-periodic digital signals are \_\_\_\_\_.

- Bit length

The most common type of connector used by the coaxial cable is \_\_\_\_\_.

- BNC

When data is sent or received using \_\_\_\_\_ the data bits are organized in a specific order, since they can only be sent or received one after another.

### Serial data transmission

\_\_\_\_\_ have established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.

- Infrared waves

In case of Frequency shift key, the difference between two frequency is represented as \_\_\_\_\_.

- 2 delta....confirm

\_\_\_\_\_ category of coaxial cable is used for thin Ethernet.

- RG-58

Cellular telephone uses \_\_\_\_\_ waves for communication.

Radio waves

\_\_\_\_\_ layer is responsible for the creating of the datagrams.

- Network

In virtual circuit approach when a frame enters a switch and when it leaves a switch its VCI always \_\_\_\_\_.

- Remain the same

According to stats, fewer than \_\_\_\_\_ crossbar points can be used at once.

- 25%

Data format used at Transport layer of TCP/IP protocol suit is called \_\_\_\_\_.

- Packet

A virtual circuit network operates on \_\_\_\_\_ layer.

- Application

If 32 bits are sent in two seconds then the bitrate for that signal is \_\_\_\_\_.

- 64 bps
- \_\_\_\_\_ normally used for long-distance data transfer.
- Both parallel and Serial data transmission

\_\_\_\_\_ is a block coding method.

- 8B\10B

TCP\IP was chosen to be the official protocol of internet \_\_\_\_\_.

- 1983

To measure the data rate \_\_\_\_\_ theoretically formula(s) were developed.

- Four

In FSK, Baud rate is less than or equal to \_\_\_\_\_ rate.

Bit rate.....confirm

\_\_\_\_\_ signal is represented by the discrete values.

- Digital

Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are called \_\_\_\_\_.

- Radio waves

Suppose a signal is amplitude then the value of decibel will be \_\_\_\_\_.

- Positive

\_\_\_\_\_ technique does not follow analog conversion.

- QAM

In Pulse Code Modulation (PCM), the sampling is dependent on \_\_\_\_\_.

- Amplitude

\_\_\_\_\_ is the sub type of Time Division Multiplexing Technique.

- Statistical TDM

Modulation of analog signal is needed if the medium is \_\_\_\_\_ in nature.

- Bandpass

Quadrature Amplitude Modulation (QAM) is the combination of \_\_\_\_\_ and \_\_\_\_\_.

ASK-FSK

There \_\_\_\_\_ components of data communication system.

5...confirm

QPSK stands for \_\_\_\_\_.

- Quadrature phase shift key

To calculate the data rate for noisy channel \_\_\_\_\_ formula is used.

- Shannon

Binary Amplitude Shift Keying(ASK) is also called as:

- On-Off Keying....confirm

Some manufacturers provide a special port called the \_\_\_\_\_.

- IrDA

To calculate the data rate for noiseless channel \_\_\_\_\_ formula is used.

- Nyquist

Two PCs and one printer are connected in a network within a room, it is an example of \_\_\_\_\_.

- LAN...confirm from net

Analog the following cables given, \_\_\_\_\_ cable provides highest bandwidth and less attenuation.

- Fiber optical

To calculate the data for noisy channel \_\_\_\_\_ formula is used.

- Shannon

In circuit switched networks we have low efficiency but minimal \_\_\_\_\_.

- Delay

Polar encoding scheme uses \_\_\_\_\_ voltage level.

- 2....confirm

TCP/IP protocol suit consists of \_\_\_\_\_ layers.

- 5.....confirm

Bandwidth in hertz is the range of \_\_\_\_\_ contained in a composite signal.

- Frequencies....confirm

\_\_\_\_\_ is an example of a host in a network.

- Computer

Set of rules to be followed for effective communication is called \_\_\_\_\_.

- Protocol

\_\_\_\_\_ is NOT an example of a connecting device.

- TCP

When an antenna transmits radio waves, they are propagated in \_\_\_\_\_.

- Only up and down direction

Which one of the following is not a bipolar encoding level?

- One

\_\_\_\_\_ and \_\_\_\_\_ are the two types of addressing in virtual circuit approach.

- Local, Global

In transmission implements \_\_\_\_\_ noise occurs from the effect of one wire on the other

- Cross talk...confirm

In computer network the stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes \_\_\_\_\_.

- Bus

In scrambling coding scheme, the number of pulses replacing the bits will be \_\_\_\_\_?

- Equal to pulses.....confirm

In the context of analog to analog signal conversation technique. "PM" Stands for \_\_\_\_\_.

- Phase modulation

\_\_\_\_\_ relatively measurement the strength of two signals.

- Decibel

Datagram approach and circuit approach are two popular approaches which lead us to \_\_\_\_\_.

- Packet switching

In virtual circuit approach when frame enters a switch and when it leaves a switch its VCI always \_\_\_\_\_.

- Remain the Same

\_\_\_\_\_ means loss of energy in signal.

Attenuation

A local telephone network use \_\_\_\_\_ network.

Circuit switched

In Amplitude Shift Keying. \_\_\_\_\_ of the signal is/are changed.

- Amplitude

In frequency domain plot, which value is plot on Y-axis?

- Amplitude....confirm

In pulse code modulation the term sampling can also be referred as \_\_\_\_\_.

- Pulse Amplitude Modulation

In analog transmission of digital data, the required bandwidth is always proportional to the signal rate expect in \_\_\_\_\_ a digital to analog conversion technique.

- ASK

WDM stands for \_\_\_\_\_.

- Wavelength Division Multiplexing

\_\_\_\_\_ does not follows analog to analog conversion.

- QAM

On a communication link, different channels are separates by unused strips of bandwidth called as \_\_\_\_\_.

Guard band

- \_\_\_\_\_ layer converts frames coming from Data Link Layer into bits and sends

Them on the transmission medium.

- Physical

Wavelength binds the \_\_\_\_\_ of a simple sine wave to the propagation speed of the medium.

Period of the frequency

book page 61

The bandwidth delay product define the number of bits that can fill the \_\_\_\_\_.

Network

The size of the packet determined by the network and \_\_\_\_\_.

Governing protocol  
To improve the performance of line coding \_\_\_\_\_ coding was.....?

- Block coding...confirm

Which one of the following is not a sampling technique?

- Neutral sampling

\_\_\_\_\_ is normally used where speed is priority in data transfer.

Hybrid transmission

In a computer network, five stations are connected to each other in such a way that each station is connected to every other station through dedicated links. This makes \_\_\_\_\_ topology.

Mesh...confirm

Some manufacturer provide a special port called the \_\_\_\_\_ port that allows a wireless keyboard to communicate with a PC.

- IrDA

Current technology supports \_\_\_\_\_ modes for propagation of light.

- Two

We quantize the sampling output into certain levels based on range of \_\_\_\_\_ and required accuracy.

- Amplitude....confirm

\_\_\_\_\_ is the Nyquist bit rate formula for noiseless channel.

- $\text{BitRate} = 2 * \text{Bandwidth} * \log_2 L$ ...confirm

A frequency is called \_\_\_\_\_, if the rate of change in sine wave Is Instantaneous.

- Infinite frequency book page 59

Which multiplexing technique involves signals composed of light beams?

- WDM....confirm

\_\_\_\_\_ is a multiplexing technique which shifts each signal to a different carrier frequency.

- FDM...confirm

Entire band in United States is regulated by \_\_\_\_\_ authorities.

- FCC

